



SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 07: Introduction to the distribution

**Lecture 01: Generalities of Multivariate
Distribution**



Cumulative Distribution Function in $\mathbb{R}^n$

Let $X = (X_1, X_2, \dots, X_n)$ be a random variable in $\mathbb{R}^n$ with joint cumulative distribution function

$$F(x_1, x_2, \dots, x_n) = \mathbb{P}(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n).$$

Then, the following properties hold:

- **Monotonicity:** F is nondecreasing in each coordinate. That is, for any $(x_1, \dots, x_n)$ and $(y_1, \dots, y_n)$ with $x_i \leq y_i$ for all i ,

$$F(x_1, \dots, x_n) \leq F(y_1, \dots, y_n).$$

- **Right-continuity:** F is right-continuous with respect to each coordinate.
- **Boundary conditions:**

$$F(x_1, \dots, x_n) \rightarrow \begin{cases} 0, & \text{if any } x_i \rightarrow -\infty, \\ 1, & \text{if all } x_i \rightarrow +\infty. \end{cases}$$

Cumulative Distribution Function in $\mathbb{R}^n$ (continued)

- **Nonnegative increment property:**

For any $a = (a_1, \dots, a_n)$ and $b = (b_1, \dots, b_n)$ with $a_i < b_i$ for all i , define

$$\Delta F(a, b) := \sum_{k=0}^n (-1)^k \sum_{\substack{I \subseteq \{1, \dots, n\} \\ |I|=k}} F(c_I),$$

where

$$(c_I)_i = \begin{cases} a_i, & \text{if } i \in I, \\ b_i, & \text{if } i \notin I. \end{cases}$$

Then,

$$\Delta F(a, b) = \mathbb{P}(a_1 < X_1 \leq b_1, \dots, a_n < X_n \leq b_n) \geq 0.$$

Definition

- An n -dimensional (or multivariate) random variable

$$X = (X_1, X_2, \dots, X_n)$$

is said to be of the **discrete type** if it takes on n -tuples of values belonging to a countable set $A \subset \mathbb{R}^n$ with probability 1. Each point

$$(x_{1i_1}, x_{2i_2}, \dots, x_{ni_n}) \in A$$

that is assumed with positive probability

$$p_{i_1 i_2 \dots i_n} = \mathbb{P}(X_1 = x_{1i_1}, X_2 = x_{2i_2}, \dots, X_n = x_{ni_n})$$

is called a *jump point* of the distribution function of X , and $p_{i_1 i_2 \dots i_n}$ is called the *jump* at that point. The set A is the *support* of the distribution of X .

Definition

- Clearly,

$$\sum_{i_1, i_2, \dots, i_n} p_{i_1 i_2 \dots i_n} = 1.$$

As for the distribution function of X , we have

$$F(x_1, x_2, \dots, x_n) = \sum_B p_{i_1 i_2 \dots i_n},$$

where

$$B = \{(i_1, i_2, \dots, i_n) : x_{1i_1} \leq x_1, x_{2i_2} \leq x_2, \dots, x_{ni_n} \leq x_n\}.$$

- The function

$$p_{i_1 i_2 \dots i_n} = \mathbb{P}(X_1 = x_{1i_1}, X_2 = x_{2i_2}, \dots, X_n = x_{ni_n}), \quad i_k = 1, 2, \dots$$

is called the **joint probability mass function (PMF)** of X .

Example

A fair die is rolled, and a fair coin is tossed independently. Let X be the face value on the die, and let $Y = 0$ if a tail turns up and $Y = 1$ if a head turns up. Then

$$A = \{(1, 0), (2, 0), \dots, (6, 0), (1, 1), (2, 1), \dots, (6, 1)\},$$

$$p_{ij} = \frac{1}{12}, \quad \text{for } i = 1, 2, \dots, 6; j = 0, 1.$$

The cdf of (X, Y) is given by

Example (continued)

$$F(x, y) = \begin{cases} 0, & x < 1, -\infty < y < \infty; \text{ or } -\infty < x < \infty, y < 0, \\ \frac{1}{12}, & 1 \leq x < 2, 0 \leq y < 1, \\ \frac{1}{6}, & 2 \leq x < 3, 0 \leq y < 1; 1 \leq x < 2, 1 \leq y, \\ \frac{1}{4}, & 3 \leq x < 4, 0 \leq y < 1, \\ \frac{1}{3}, & 4 \leq x < 5, 0 \leq y < 1; 2 \leq x < 3, 1 \leq y, \\ \frac{5}{12}, & 5 \leq x < 6, 0 \leq y < 1, \\ \frac{1}{2}, & 6 \leq x, 0 \leq y < 1; 3 \leq x < 4, 1 \leq y, \\ \frac{2}{3}, & 4 \leq x < 5, 1 \leq y, \\ \frac{5}{6}, & 5 \leq x < 6, 1 \leq y, \\ 1, & 6 \leq x, 1 \leq y. \end{cases}$$

Definition (continued)

An n -dimensional random variable

$$X = (X_1, X_2, \dots, X_n)$$

is said to be of the **continuous type** if there exists a nonnegative function $f(\cdot)$ such that, for every $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$, we have

$$F(x_1, x_2, \dots, x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(u_1, u_2, \dots, u_n) du_n \cdots du_1,$$

where F is the cdf of X . The function f is called the **(joint) probability density function (PDF)** of X .

Clearly,

$$\begin{aligned} F(+\infty, +\infty, \dots, +\infty) &= \lim_{x_1, \dots, x_n \rightarrow +\infty} \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(u_1, u_2, \dots, u_n) du_n \cdots du_1 \\ &= \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(u_1, u_2, \dots, u_n) du_n \cdots du_1 = 1. \end{aligned}$$

Definitions

Let

$$X = (X_1, X_2, \dots, X_n)$$

be an n -dimensional discrete random variable with joint PMF

$$p_{i_1 i_2 \dots i_n} = \mathbb{P}\{X_1 = x_{1i_1}, X_2 = x_{2i_2}, \dots, X_n = x_{ni_n}\}.$$

Then, for any fixed $k \in \{1, 2, \dots, n\}$, the marginal probability mass function of X_k is obtained by summing the joint PMF over all other indices:

$$\mathbb{P}\{X_k = x_{ki_k}\} = \sum_{\substack{i_1, \dots, i_n \\ i_j \text{ for } j \neq k}} p_{i_1 i_2 \dots i_n}.$$

We denote this by

$$p_{i_k \cdot} := \sum_{\substack{i_1, \dots, i_n \\ i_j \text{ for } j \neq k}} p_{i_1 i_2 \dots i_n}.$$

Clearly, $p_{i_k \cdot} \geq 0$ and

$$\sum_{i_k=1}^{\infty} p_{i_k \cdot} = 1.$$

The collection $\{p_{i_k \cdot}\}$ represents the **marginal PMF** of X_k .

Example

A fair coin is tossed three times. Let X = number of heads in three tossings, and Y = difference, in absolute value, between number of heads and number of tails. The joint PMF of (X, Y) is given in the following table:

$Y \backslash X$	0	1	2	3	$P(Y = y)$
1	0	$\frac{3}{8}$	$\frac{3}{8}$	0	$\frac{3}{4}$
3	$\frac{1}{8}$	0	0	$\frac{1}{8}$	$\frac{1}{4}$
$P(X = x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	1

The marginal PMF of Y is shown in the column representing row totals, and the marginal PMF of X , in the row representing column totals.

Definition

If

$$X = (X_1, X_2, \dots, X_n)$$

is a random variable of the **continuous type** with joint PDF

$$f(x_1, x_2, \dots, x_n),$$

then, for any fixed $k \in \{1, 2, \dots, n\}$, the **marginal PDF** of X_k is obtained by integrating out all other variables:

$$f_k(x_k) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_1 \cdots dx_{k-1} dx_{k+1} \cdots dx_n.$$

Clearly, $f_k(x_k) \geq 0$ and

$$\int_{-\infty}^{\infty} f_k(x_k) dx_k = 1.$$

Hence, each $f_k(x_k)$ is a valid PDF and represents the **marginal PDF of X_k** .

More generally, for any subset of indices $S \subseteq \{1, 2, \dots, n\}$, the joint marginal PDF of $\{X_i : i \in S\}$ is obtained by integrating f over all variables not in S .

Example

Let (X, Y) be jointly distributed with PDF

$$f(x, y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then

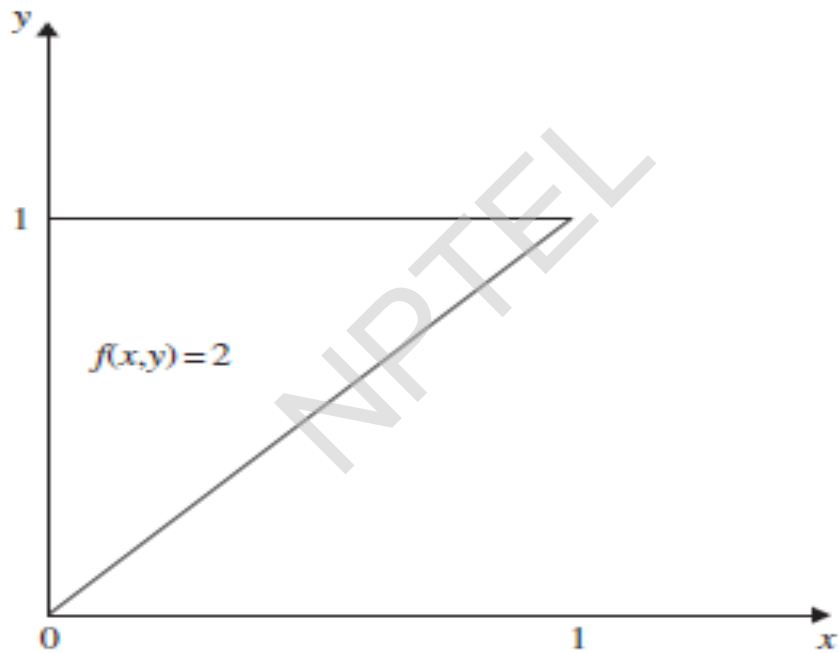
$$f_1(x) = \int_x^1 2 \, dy = \begin{cases} 2 - 2x, & 0 < x < 1, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$f_2(y) = \int_0^y 2 \, dx = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

These are the two marginal density functions.

Example (continued)



Definition

Definition (Multivariate Conditional PMF / PDF). Let

$$X = (X_1, X_2, \dots, X_m), \quad Y = (Y_1, Y_2, \dots, Y_n)$$

be jointly distributed random variables.

(a) Discrete case: If $P\{Y = y_j\} > 0$, then the *conditional PMF of X given $Y = y_j$* is

$$P\{X = x_i \mid Y = y_j\} = \frac{P\{X = x_i, Y = y_j\}}{P\{Y = y_j\}}.$$

(b) Continuous case: If $f_Y(y) > 0$, then the *conditional PDF of X given $Y = y$* is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)},$$

where $f_{X,Y}$ is the joint PDF of (X, Y) and f_Y is the marginal PDF of Y .

Example

For the joint PMF of coin toss example , we have for $Y = 1$:

$$P\{X = i \mid Y = 1\} = \begin{cases} 0, & i = 0, 3, \\ \frac{1}{2}, & i = 1, 2. \end{cases}$$

Similarly,

$$P\{X = i \mid Y = 3\} = \begin{cases} \frac{1}{2}, & i = 0, 3, \\ 0, & i = 1, 2, \end{cases} \quad P\{Y = j \mid X = 0\} = \begin{cases} 0, & j = 1, \\ 1, & j = 3. \end{cases}$$

and so on.

Definition

Definition (Conditional Expectation in the Multivariate Case).

Let (X, Y) be jointly distributed random variables, where

$$X = (X_1, X_2, \dots, X_m), \quad Y = (Y_1, Y_2, \dots, Y_n).$$

Then the **conditional expectation** of X given $Y = y$ is defined as

$$E[X \mid Y = y] = \begin{cases} \sum x P(X = x \mid Y = y), & \text{if discrete,} \\ \int_{\mathbb{R}^m} x f_{X|Y}(x|y) dx, & \text{if continuous.} \end{cases}$$

The function $g(y) = E[X \mid Y = y]$ defines a random variable

$$E[X \mid Y] = g(Y),$$

which is called the **conditional expectation of X given Y** .

Properties of Conditional Expectation (Multivariate Case)

For random vectors $X, Z \in \mathbb{R}^m$ and $Y \in \mathbb{R}^n$:

- **Linearity:**

$$E[aX + bZ \mid Y] = aE[X \mid Y] + bE[Z \mid Y], \quad a, b \in \mathbb{R}.$$

- **Law of total expectation:**

$$E[E[X \mid Y]] = E[X].$$

- **Independence:** If X and Y are independent,

$$E[X \mid Y] = E[X].$$

- **Function of conditioning variable:**

$$E[h(Y) \mid Y] = h(Y) \quad \text{for any measurable function } h.$$

- **Tower property (iterated conditioning):** For another random vector Z ,

$$E[E[X \mid Y, Z] \mid Y] = E[X \mid Y].$$

Example: Discrete Case

Suppose a fair coin is tossed twice. Let:

$$X = \text{number of heads}, \quad Y = \begin{cases} 1, & \text{if first toss is head,} \\ 0, & \text{if first toss is tail.} \end{cases}$$

Then:

$$E[X \mid Y = 1] = 1.5, \quad E[X \mid Y = 0] = 0.5.$$

$$E[X] = E[E[X \mid Y]] = (1.5)(0.5) + (0.5)(0.5) = 1.$$

Continuous case

Let the joint PDF of random variables X and Y be

$$f(x, y) = \begin{cases} 2, & 0 < x < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

The marginal PDF of Y is

$$f_Y(y) = \int_0^y 2 \, dx = 2y, \quad 0 < y < 1.$$

The conditional PDF of X given $Y = y$ is

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)} = \frac{2}{2y} = \frac{1}{y}, \quad 0 < x < y.$$

Example: Continuous case

Thus, $X|Y = y$ is uniformly distributed on $(0, y)$.

$$E[X | Y = y] = \int_0^y x \cdot \frac{1}{y} dx = \frac{y}{2}.$$

Hence, as a random variable,

$$E[X | Y] = \frac{Y}{2}.$$

Example (continued)

By the law of total expectation,

$$E[X] = E[E[X \mid Y]] = E\left[\frac{Y}{2}\right] = \frac{1}{2}E[Y].$$

Compute $E[Y]$:

$$E[Y] = \int_0^1 y(2y) dy = 2 \int_0^1 y^2 dy = \frac{2}{3}.$$

Therefore,

$$E[X] = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

Conclusion: $E[X \mid Y] = \frac{Y}{2}$ and $E[X] = \frac{1}{3}$.

References

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- ② M. H. DeGroot and M. J. Schervish, Probability and Statistics, 4th ed., Pearson.
- ③ W. Feller, An Introduction to Probability Theory and Its Applications, Vol. 1, 3rd ed., Wiley

Thank you





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Module 07: Introduction to the distribution

Lecture 02: Bivariate Normal Distribution



Introduction

The **bivariate normal distribution** models two continuous random variables (X, Y) that follow a joint Gaussian law. It generalizes the univariate normal distribution to two dimensions, capturing both individual variation and linear dependence between variables.

Real-Life Motivation

- Height and weight of individuals.
- Length and width of leaves.
- Returns on two correlated stocks.
- Exam scores in two related subjects.

The bivariate normal captures both individual normality and the correlation structure between the two variables.

Definition

A random vector (X, Y) is said to have a **bivariate normal distribution** with parameters $\mu_X, \mu_Y, \sigma_X, \sigma_Y$, and ρ if its joint probability density function (PDF) is given by:

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} \left(\frac{(x-\mu_X)^2}{\sigma_X^2} - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right)\right].$$

where μ_X, μ_Y are real numbers, σ_X, σ_Y are non-negative real numbers, and $|\rho| \leq 1$.

Bivariate Central Limit Theorem

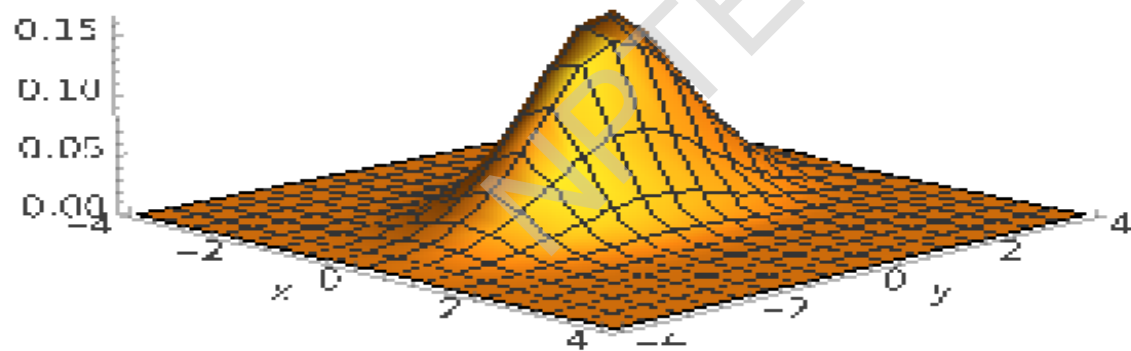
For large n , the average of “any” i.i.d. bivariate sample of size n follows bivariate normal distribution.

Parameters

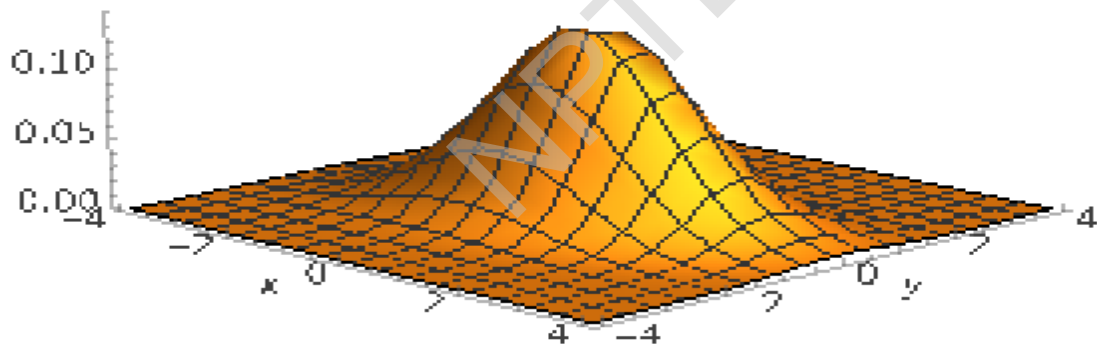
- μ_X, μ_Y : means of X and Y .
- σ_X, σ_Y : standard deviations.
- ρ : correlation coefficient.

These parameters completely determine the shape, spread, and orientation of the joint distribution.

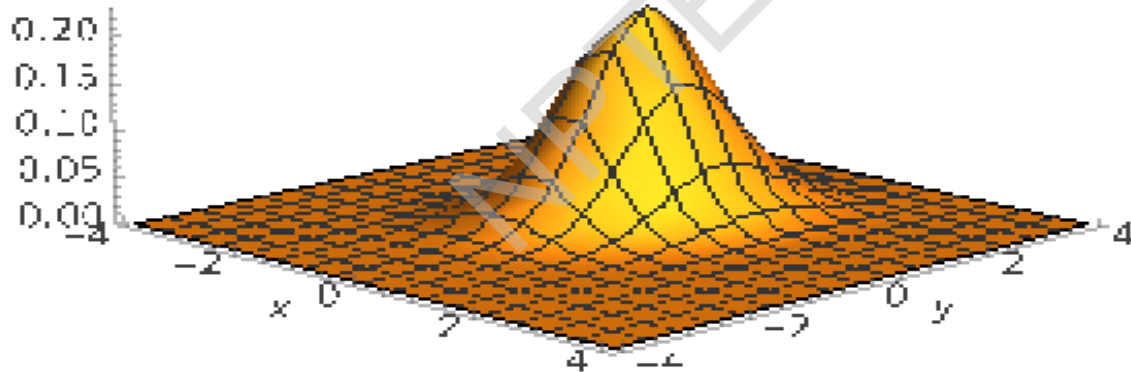
μ_X	μ_Y	σ_X	σ_Y	ρ
-0.39	0.31	0.736	1.36	-0.326



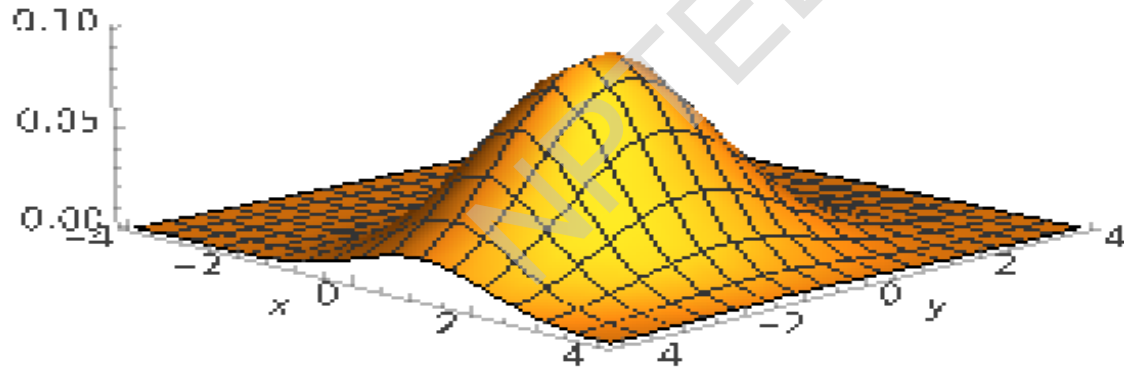
μ_X	μ_Y	σ_X	σ_Y	ρ
0.365	-0.335	1.396	0.82	0.214



μ_X	μ_Y	σ_X	σ_Y	ρ
-0.145	0.58	0.71	0.972	0.124



μ_X	μ_Y	σ_X	σ_Y	ρ
1.085	-1.255	1.042	1.498	0.06



Covariance Matrix

The covariance matrix of (X, Y) is

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}.$$

It encodes both the individual variances and the covariance (linear dependence) between X and Y .

Marginal Distributions

The marginal distributions of X and Y are both normal:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2).$$

Even though X and Y are jointly distributed, their individual behaviors remain normal.

Conditional Distributions

For the bivariate normal distribution,

$$X \mid Y = y \sim \mathcal{N}\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y}(y - \mu_Y), \sigma_X^2(1 - \rho^2)\right).$$

Similarly,

$$Y \mid X = x \sim \mathcal{N}\left(\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X), \sigma_Y^2(1 - \rho^2)\right).$$

Interpretation of Conditional Expectation

The conditional expectation $E[X|Y = y]$ is a linear function of y :

$$E[X|Y = y] = \mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y).$$

This expresses the best linear prediction of X based on the observed value of Y .

Special Case: Independence

If $\rho = 0$, then

$$f(x, y) = f_X(x)f_Y(y),$$

and X and Y are independent. The joint density contours are circular (if $\sigma_X = \sigma_Y$) or axis-aligned ellipses otherwise.

Geometry of the Distribution

The joint density of (X, Y) forms **elliptical contours**:

- $\rho > 0$: Ellipses tilt upward (positive correlation).
- $\rho < 0$: Ellipses tilt downward (negative correlation).
- $\rho = 0$: Circles or axis-aligned ellipses (independence).

Example 1: Independent Case

Let $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$, and $\rho = 0$. Then

$$f(x, y) = \frac{1}{2\pi} e^{-\frac{1}{2}(x^2 + y^2)}.$$

X and Y are independent standard normal random variables.

Example 2: Correlated Case

Let $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$, and $\rho = 0.8$. Then

$$f(x, y) = \frac{1}{2\pi\sqrt{1-0.8^2}} \exp\left[-\frac{1}{2(1-0.8^2)}(x^2 - 1.6xy + y^2)\right].$$

X and Y are strongly positively correlated; most probability mass lies near the line $y = x$.

Example 3: Negative Correlation

If $\rho = -0.8$, the density becomes concentrated near $y = -x$. The conditional means are negatively related:

$$E[X|Y = y] = -0.8y.$$

This corresponds to downward-tilted ellipses in the joint PDF.

Important Properties

- Marginals and conditionals are normal.
- $E[X] = \mu_X$, $E[Y] = \mu_Y$.
- $\text{Cov}(X, Y) = \rho\sigma_X\sigma_Y$.
- $E[X|Y]$ and $E[Y|X]$ are linear.

The bivariate normal is the only joint distribution where linear correlation fully describes dependence.

Bivariate Normal Distribution

For random variables X, Y with means μ_X, μ_Y , standard deviations σ_X, σ_Y , and correlation ρ :

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{1}{2(1-\rho^2)} Q(x, y)\right]$$

where

$$Q(x, y) = \left(\frac{x - \mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x - \mu_X}{\sigma_X}\right)\left(\frac{y - \mu_Y}{\sigma_Y}\right) + \left(\frac{y - \mu_Y}{\sigma_Y}\right)^2$$

Contours of Constant Density

For a fixed constant $c > 0$, the set of points satisfying

$$Q(x, y) = c$$

forms a contour of equal probability density. This contour is an **ellipse** centered at (μ_X, μ_Y) . It represents all points having the same likelihood under the bivariate normal distribution.

Geometric Interpretation

A general second-degree (quadratic) curve in variables x and y can be written as

$$Ax^2 + 2Bxy + Cy^2 = \text{constant}.$$

The nature of the curve depends on the discriminant $\Delta = B^2 - AC$:

$$\begin{cases} \Delta < 0 & \text{Ellipse,} \\ \Delta = 0 & \text{Parabola,} \\ \Delta > 0 & \text{Hyperbola.} \end{cases}$$

For the bivariate normal contour, we have

$$A = \frac{1}{\sigma_X^2}, \quad B = -\frac{\rho}{\sigma_X \sigma_Y}, \quad C = \frac{1}{\sigma_Y^2}.$$

Hence,

$$B^2 - AC = \frac{\rho^2}{\sigma_X^2 \sigma_Y^2} - \frac{1}{\sigma_X^2 \sigma_Y^2} = \frac{\rho^2 - 1}{\sigma_X^2 \sigma_Y^2} < 0,$$

since $|\rho| < 1$.

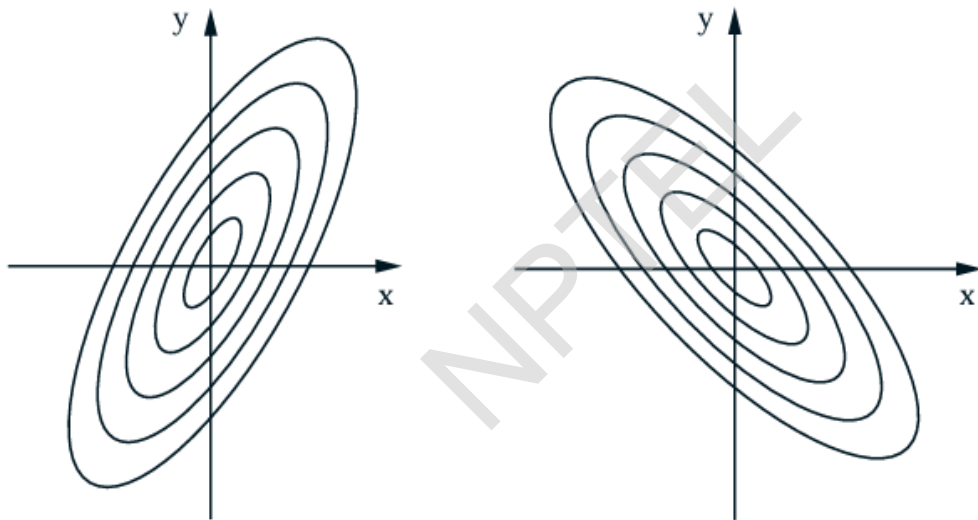
Therefore, the constant-density contours of the bivariate normal distribution are **ellipses**.

Effect of Correlation

- $\rho = 0$: ellipse axes align with coordinate axes.
- $\rho > 0$: ellipse tilts upward (positive association).
- $\rho < 0$: ellipse tilts downward (negative association).

The correlation ρ controls the **orientation** and **elongation** of the ellipse.

A picture of contour ellipse



Contours of the bivariate normal PDF. The diagram on the left (respectively, right) corresponds to a case of positive (respectively, negative) correlation coefficient ρ .

Covariance Matrix and Axes

$$\Sigma = \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}$$

- Eigenvectors of Σ give the **directions** of the ellipse axes.
- Eigenvalues λ_1, λ_2 determine the squared lengths:

Semi-axes lengths $\propto \sqrt{\lambda_1}, \sqrt{\lambda_2}$.

Angle of Rotation

The angle θ (tilt of the ellipse with respect to the x -axis) is

$$\tan(2\theta) = \frac{2\rho\sigma_X\sigma_Y}{\sigma_X^2 - \sigma_Y^2}.$$

This shows how the correlation ρ determines the **rotation of the ellipse** in the xy -plane.

Summary

- The contours of a bivariate normal distribution are **ellipses**.
- These ellipses represent **constant probability density**.
- The shape, size, and orientation of ellipses depend on the covariance matrix.

Joint PDF (setup)

Let (X, Y) have joint PDF

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}Q(x, y)\right\},$$

where

$$Q(x, y) = \frac{(x - \mu_X)^2}{\sigma_X^2} - 2\rho\frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X\sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2}.$$

We will prove: (A) marginals are normal, (B) conditional $X|Y = y$ is normal (give mean and var), (C) $E[X] = \mu_X$, $\text{Cov}(X, Y) = \rho\sigma_X\sigma_Y$, (D) $\rho = 0$ implies independence.

Marginal of X : integrate out y

Compute

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2(1-\rho^2)}Q(x, y)\right\} dy.$$

To evaluate the integral, view x fixed and treat $Q(x, y)$ as a quadratic in y and complete the square.

Complete the square in y (details)

Start with the y -part of $Q(x, y)$:

$$Q(x, y) = \frac{(y - \mu_Y)^2}{\sigma_Y^2} - 2\rho \frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(x - \mu_X)^2}{\sigma_X^2}.$$

Focus on the first two terms and rewrite:

$$\frac{(y - \mu_Y)^2}{\sigma_Y^2} - 2\rho \frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} = \frac{1}{\sigma_Y^2} \left((y - \mu_Y)^2 - 2\rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)(y - \mu_Y) \right).$$

Complete the square inside the bracket:

$$(y - \mu_Y)^2 - 2\rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)(y - \mu_Y) = \left((y - \mu_Y) - \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \right)^2 - \rho^2 \frac{\sigma_Y^2}{\sigma_X^2} (x - \mu_X)^2.$$

Therefore

$$Q(x, y) = \frac{1}{\sigma_Y^2} \left((y - \mu_Y) - \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \right)^2 + \left(1 - \rho^2 \right) \frac{(x - \mu_X)^2}{\sigma_X^2}.$$

Finish integration for $f_X(x)$

Plugging the expression for $Q(x, y)$ into $f_X(x)$:

$$f_X(x) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2}\frac{(x-\mu_X)^2}{\sigma_X^2}\right\} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2(1-\rho^2)} \cdot \frac{1}{\sigma_Y^2} \left((y-\mu_Y) - \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)\right)^2\right\} dy.$$

Change variable: let

$$u = \frac{(y-\mu_Y) - \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)}{\sigma_Y\sqrt{1-\rho^2}},$$

then $dy = \sigma_Y\sqrt{1-\rho^2} du$. The integral becomes

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 1.$$

Thus

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp\left\{-\frac{(x-\mu_X)^2}{2\sigma_X^2}\right\},$$

so $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$.

Conditional PDF $f_{X|Y}(x|y)$

By definition

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

To get its form, either compute $f_Y(y)$ by integrating out x (symmetric to above) or isolate the x dependence in $f(x, y)$ by completing the square in x . We proceed to complete the square in x (fixing y).

Complete the square in x (details)

Write the quadratic form $Q(x, y)$ grouping x terms:

$$Q(x, y) = \frac{(x - \mu_X)^2}{\sigma_X^2} - 2\rho \frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2}.$$

Factor the x -dependent part:

$$\frac{1}{\sigma_X^2} \left((x - \mu_X)^2 - 2\rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)(x - \mu_X) \right).$$

Complete the square:

$$(x - \mu_X)^2 - 2\rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)(x - \mu_X) = \left((x - \mu_X) - \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y) \right)^2 - \rho^2 \frac{\sigma_X^2}{\sigma_Y^2} (y - \mu_Y)^2.$$

Hence

$$Q(x, y) = \frac{1}{\sigma_X^2} \left((x - \mu_X) - \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y) \right)^2 + (1 - \rho^2) \frac{(y - \mu_Y)^2}{\sigma_Y^2}.$$

Form of $f_{X|Y}(x|y)$

From the completed square,

$$f(x, y) = \underbrace{\frac{1}{\sqrt{2\pi}\sigma_X\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)} \cdot \frac{1}{\sigma_X^2} \left((x - \mu_X) - \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)\right)^2\right\}}_{\text{term depending on } x} \cdot \underbrace{\frac{1}{\sqrt{2\pi}\sigma_Y} \exp\left\{-\frac{1}{2} \frac{(y - \mu_Y)^2}{\sigma_Y^2}\right\}}_{\text{term depending on } y}.$$

Thus the factor multiplying the y -term is exactly $f_{X|Y}(x|y)$ up to normalization. Dividing by $f_Y(y)$ yields

$$f_{X|Y}(x|y) = \frac{1}{\sqrt{2\pi}\sigma_X\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2} \frac{(x - (\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)))^2}{\sigma_X^2(1-\rho^2)}\right\}.$$

Therefore

$$X | Y = y \sim \mathcal{N}\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y), \sigma_X^2(1-\rho^2)\right).$$

Means: $E[X] = \mu_X$, $E[Y] = \mu_Y$

Because $f_X(x)$ is $\mathcal{N}(\mu_X, \sigma_X^2)$, we have directly

$$E[X] = \mu_X, \quad \text{Var}(X) = \sigma_X^2.$$

Similarly $f_Y(y)$ is $\mathcal{N}(\mu_Y, \sigma_Y^2)$, so $E[Y] = \mu_Y$.

Covariance: compute $\text{Cov}(X, Y)$

Use the law of total expectation:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[E[(X - \mu_X)(Y - \mu_Y) | Y]].$$

Inside the conditional expectation, treat Y fixed:

$$E[(X - \mu_X)(Y - \mu_Y) | Y = y] = (y - \mu_Y) E[X - \mu_X | Y = y].$$

From the conditional mean found earlier,

$$E[X - \mu_X | Y = y] = \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y).$$

Hence

$$E[(X - \mu_X)(Y - \mu_Y) | Y = y] = \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)^2.$$

Now take expectation in Y :

$$\text{Cov}(X, Y) = \rho \frac{\sigma_X}{\sigma_Y} E[(Y - \mu_Y)^2] = \rho \frac{\sigma_X}{\sigma_Y} \sigma_Y^2 = \rho \sigma_X \sigma_Y.$$

Independence when $\rho = 0$

If $\rho = 0$, the joint PDF simplifies:

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y} \exp\left\{-\frac{1}{2}\left(\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right)\right\}.$$

This factors as

$$f(x, y) = \underbrace{\frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}}_{f_X(x)} \cdot \underbrace{\frac{1}{\sqrt{2\pi}\sigma_Y} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}}}_{f_Y(y)}.$$

Since $f(x, y) = f_X(x)f_Y(y)$ for all x, y , X and Y are independent.

Properties of Bivariate Normal Distribution

- X and Y are independent if and only if they are uncorrelated.
- The conditional expectation of X given Y satisfies

$$\mathbb{E}[X | Y] = \mathbb{E}[X] + \rho \frac{\sigma_X}{\sigma_Y} (Y - \mathbb{E}[Y]).$$

It is a linear function of Y and has a normal PDF.

- The estimation error $\tilde{X} = X - \mathbb{E}[X | Y]$ is zero-mean, normal, and independent of Y , with variance

$$\sigma_{\tilde{X}}^2 = (1 - \rho^2) \sigma_X^2.$$

- The conditional distribution of X given Y is normal with mean $\mathbb{E}[X | Y]$ and variance $\sigma_{\tilde{X}}^2$.
- If (X, Y) has a bivariate normal distribution, then any linear combination of X and Y is a normal distribution.

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Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 07: Introduction to Multivariate Distribution

**Lecture 03: Multivariate Normal
Distribution 1**



PDF of Bivariate Normal Distribution

Let (X, Y) have joint PDF

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)} Q(x, y)\right\},$$

where

$$Q(x, y) = \frac{(x - \mu_X)^2}{\sigma_X^2} - 2\rho \frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X\sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2}.$$

- Marginals are normal.
- Conditional $X|Y = y$ is normal.
- $E[X] = \mu_X$, and $E[Y] = \mu_Y$
- $\text{Var}(X) = \sigma_X^2$ and $\text{Var}(Y) = \sigma_Y^2$
- $\text{Cov}(X, Y) = \rho\sigma_X\sigma_Y$.
- $\rho = 0$ implies independence.

Marginal of X : integrate out y

Compute

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2(1-\rho^2)}Q(x, y)\right\} dy.$$

To evaluate the integral, view x fixed and treat $Q(x, y)$ as a quadratic in y and complete the square.

Complete the square in y (details)

Start with the y -part of $Q(x, y)$:

$$Q(x, y) = \frac{(y - \mu_Y)^2}{\sigma_Y^2} - 2\rho \frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(x - \mu_X)^2}{\sigma_X^2}.$$

Focus on the first two terms and rewrite:

$$\frac{(y - \mu_Y)^2}{\sigma_Y^2} - 2\rho \frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} = \frac{1}{\sigma_Y^2} \left((y - \mu_Y)^2 - 2\rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)(y - \mu_Y) \right).$$

Complete the square inside the bracket:

$$(y - \mu_Y)^2 - 2\rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X)(y - \mu_Y) = \left((y - \mu_Y) - \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \right)^2 - \rho^2 \frac{\sigma_Y^2}{\sigma_X^2} (x - \mu_X)^2.$$

Therefore

$$Q(x, y) = \frac{1}{\sigma_Y^2} \left((y - \mu_Y) - \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \right)^2 + \left(1 - \rho^2 \right) \frac{(x - \mu_X)^2}{\sigma_X^2}.$$

Finish integration for $f_X(x)$

Plugging the expression for $Q(x, y)$ into $f_X(x)$:

$$f_X(x) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2}\frac{(x-\mu_X)^2}{\sigma_X^2}\right\} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2(1-\rho^2)} \cdot \frac{1}{\sigma_Y^2} \left((y-\mu_Y) - \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)\right)^2\right\} dy.$$

Change variable: let

$$u = \frac{(y-\mu_Y) - \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)}{\sigma_Y\sqrt{1-\rho^2}},$$

then $dy = \sigma_Y\sqrt{1-\rho^2} du$. The integral becomes

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = 1.$$

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$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp\left\{-\frac{(x-\mu_X)^2}{2\sigma_X^2}\right\},$$

so $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$.

Conditional PDF $f_{X|Y}(x|y)$

By definition

$$f_{X|Y}(x|y) = \frac{f(x, y)}{f_Y(y)}.$$

To get its form, either compute $f_Y(y)$ by integrating out x (symmetric to above) or isolate the x dependence in $f(x, y)$ by completing the square in x . We proceed to complete the square in x (fixing y).

Complete the square in x (details)

Write the quadratic form $Q(x, y)$ grouping x terms:

$$Q(x, y) = \frac{(x - \mu_X)^2}{\sigma_X^2} - 2\rho \frac{(x - \mu_X)(y - \mu_Y)}{\sigma_X \sigma_Y} + \frac{(y - \mu_Y)^2}{\sigma_Y^2}.$$

Factor the x -dependent part:

$$\frac{1}{\sigma_X^2} \left((x - \mu_X)^2 - 2\rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)(x - \mu_X) \right).$$

Complete the square:

$$(x - \mu_X)^2 - 2\rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)(x - \mu_X) = \left((x - \mu_X) - \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y) \right)^2 - \rho^2 \frac{\sigma_X^2}{\sigma_Y^2} (y - \mu_Y)^2.$$

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Form of $f_{X|Y}(x|y)$

From the completed square,

$$f(x, y) = \underbrace{\frac{1}{\sqrt{2\pi}\sigma_X\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)} \cdot \frac{1}{\sigma_X^2} \left((x - \mu_X) - \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)\right)^2\right\}}_{\text{term depending on } x} \cdot \underbrace{\frac{1}{\sqrt{2\pi}\sigma_Y} \exp\left\{-\frac{1}{2} \frac{(y - \mu_Y)^2}{\sigma_Y^2}\right\}}_{\text{term depending on } y}.$$

Thus the factor multiplying the y -term is exactly $f_{X|Y}(x|y)$ up to normalization. Dividing by $f_Y(y)$ yields

$$f_{X|Y}(x|y) = \frac{1}{\sqrt{2\pi}\sigma_X\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2} \frac{(x - (\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)))^2}{\sigma_X^2(1-\rho^2)}\right\}.$$

Therefore

$$X | Y = y \sim \mathcal{N}\left(\mu_X + \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y), \sigma_X^2(1-\rho^2)\right).$$

Means: $E[X] = \mu_X$, $E[Y] = \mu_Y$

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$$E[X] = \mu_X, \quad \text{Var}(X) = \sigma_X^2.$$

Similarly $f_Y(y)$ is $\mathcal{N}(\mu_Y, \sigma_Y^2)$, so $E[Y] = \mu_Y$.

Covariance: compute $\text{Cov}(X, Y)$

Use the law of total expectation:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[E[(X - \mu_X)(Y - \mu_Y) | Y]].$$

Inside the conditional expectation, treat Y fixed:

$$E[(X - \mu_X)(Y - \mu_Y) | Y = y] = (y - \mu_Y) E[X - \mu_X | Y = y].$$

From the conditional mean found earlier,

$$E[X - \mu_X | Y = y] = \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y).$$

Hence

$$E[(X - \mu_X)(Y - \mu_Y) | Y = y] = \rho \frac{\sigma_X}{\sigma_Y} (y - \mu_Y)^2.$$

Now take expectation in Y :

$$\text{Cov}(X, Y) = \rho \frac{\sigma_X}{\sigma_Y} E[(Y - \mu_Y)^2] = \rho \frac{\sigma_X}{\sigma_Y} \sigma_Y^2 = \rho \sigma_X \sigma_Y.$$

Independence when $\rho = 0$

If $\rho = 0$, the joint PDF simplifies:

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y} \exp\left\{-\frac{1}{2}\left(\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}\right)\right\}.$$

This factors as

$$f(x, y) = \underbrace{\frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}}_{f_X(x)} \cdot \underbrace{\frac{1}{\sqrt{2\pi}\sigma_Y} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}}}_{f_Y(y)}.$$

Since $f(x, y) = f_X(x)f_Y(y)$ for all x, y , X and Y are independent.

Properties of Bivariate Normal Distribution

- X and Y are independent if and only if they are uncorrelated.
- The conditional expectation of X given Y satisfies

$$\mathbb{E}[X | Y] = \mathbb{E}[X] + \rho \frac{\sigma_X}{\sigma_Y} (Y - \mathbb{E}[Y]).$$

It is a linear function of Y and has a normal PDF.

- The estimation error $\tilde{X} = X - \mathbb{E}[X | Y]$ is zero-mean, normal, and independent of Y , with variance

$$\sigma_{\tilde{X}}^2 = (1 - \rho^2) \sigma_X^2.$$

- The conditional distribution of X given Y is normal with mean $\mathbb{E}[X | Y]$ and variance $\sigma_{\tilde{X}}^2$.
- If (X, Y) has a bivariate normal distribution, then any linear combination of X and Y is a normal distribution.

Multivariate Normal Distribution

A vector $X = (X_1, \dots, X_p)^\top$ is said to follow multivariate normal distribution if all the linear combinations $\sum_{i=1}^p a_i X_i$ follow univariate normal distribution.

PDF of Multivariate Normal Distribution

Let X be a $p \times 1$ random vector and μ represent the expected value of X . Let Σ be the variance-covariance matrix of order $p \times p$, (we shall assume that Σ is a positive definite matrix); then the probability distribution function for the multivariate normal distribution for the random vector $X = [X_1, \dots, X_p]^\top$ is,

$$f(x) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} e^{-\{(x-\mu)^\top \Sigma^{-1} (x-\mu)/2\}} \quad -\infty < x_i < \infty, i = 1, 2, \dots, p.$$

We shall denote this p - dimensional normal density by $\mathcal{N}_p(\mu, \Sigma)$.

One Variable Case

- Recall that the pdf of one variable normal distribution is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\{(x-\mu)^2/2\sigma^2\}}, \quad -\infty < x < \infty$$

- Observe that we can write $\left(\frac{x-\mu}{\sigma}\right)^2 = (x-\mu)(\sigma^2)^{-1}(x-\mu)$

Example

The following is a four variate normal PDF,

$$\frac{1}{48\pi^2} \exp \left\{ -\frac{1}{2} \left[\frac{(x_1 - 2)^2}{6} + \frac{(x_2 + 1)^2}{3} + \frac{x_3^2}{4} + \frac{(x_4 - 4)^2}{2} \right] \right\}.$$

Example (continued)

Observe that the given 4-variate normal distribution does not have any correlation terms. So, we compare it with the following density function:

$$\frac{1}{(2\pi)^{4/2} \prod_{i=1}^4 \sigma_i} \exp \left\{ -\frac{1}{2} \left[\frac{(x_1 - \mu_1)^2}{\sigma_1^2} + \frac{(x_2 - \mu_2)^2}{\sigma_2^2} + \frac{(x_3 - \mu_3)^2}{\sigma_3^2} + \frac{(x_4 - \mu_4)^2}{\sigma_4^2} \right] \right\}.$$

Rewriting the given pdf

$$\frac{1}{(2\pi)^{4/2} \sqrt{6}\sqrt{3} \times 2 \times \sqrt{2}} \exp \left\{ -\frac{1}{2} \left[\frac{(x_1 - 2)^2}{6} + \frac{(x_2 + 1)^2}{3} + \frac{x_3^2}{4} + \frac{(x_4 - 4)^2}{2} \right] \right\}.$$

On comparing, we get:

$$\sigma_1 = \sqrt{6}, \quad \sigma_2 = \sqrt{3}, \quad \sigma_3 = 2, \quad \sigma_4 = \sqrt{2}.$$

$$\prod_{i=1}^4 \sigma_i = 12.$$

Example (continued)

Hence, the mean vector and the variance-covariance matrix of the distribution are given by:

$$\mu = \begin{bmatrix} 2 \\ -1 \\ 0 \\ 4 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}.$$

Example (Bivariate normal density)

Let us evaluate the 2-variate normal density in terms of the individual parameters $\mu_1 = E(X_1)$, $\mu_2 = E(X_2)$, $\sigma_{11} = \text{Var}(X_1)$, $\sigma_{22} = \text{Var}(X_2)$, and $\rho_{12} = \sigma_{12}/(\sqrt{\sigma_{11}\sigma_{22}}) = \text{Corr}(X_1, X_2)$.

Using the covariance matrix

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{22} \end{bmatrix},$$

we find that its inverse is

$$\Sigma^{-1} = \frac{1}{\sigma_{11}\sigma_{22} - \sigma_{12}^2} \begin{bmatrix} \sigma_{22} & -\sigma_{12} \\ -\sigma_{12} & \sigma_{11} \end{bmatrix}.$$

Introducing the correlation coefficient ρ_{12} by writing $\sigma_{12} = \rho_{12}\sqrt{\sigma_{11}\sigma_{22}}$, we obtain $\sigma_{11}\sigma_{22} - \sigma_{12}^2 = \sigma_{11}\sigma_{22}(1 - \rho_{12}^2)$, and the squared distance becomes

Example (continued)

$$\begin{aligned}(x - \mu)^\top \Sigma^{-1} (x - \mu) &= [x_1 - \mu_1, x_2 - \mu_2] \frac{1}{\sigma_{11}\sigma_{22}(1 - \rho_{12}^2)} \begin{bmatrix} \sigma_{22} & -\rho_{12}\sqrt{\sigma_{11}\sigma_{22}} \\ -\rho_{12}\sqrt{\sigma_{11}\sigma_{22}} & \sigma_{11} \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \end{bmatrix} \\&= \frac{1}{\sigma_{11}\sigma_{22}(1 - \rho_{12}^2)} [\sigma_{22}(x_1 - \mu_1)^2 + \sigma_{11}(x_2 - \mu_2)^2 - 2\rho_{12}\sqrt{\sigma_{11}\sigma_{22}}(x_1 - \mu_1)(x_2 - \mu_2)] \\&= \frac{1}{1 - \rho_{12}^2} \left[\left(\frac{x_1 - \mu_1}{\sqrt{\sigma_{11}}} \right)^2 + \left(\frac{x_2 - \mu_2}{\sqrt{\sigma_{22}}} \right)^2 - 2\rho_{12} \left(\frac{x_1 - \mu_1}{\sqrt{\sigma_{11}}} \right) \left(\frac{x_2 - \mu_2}{\sqrt{\sigma_{22}}} \right) \right].\end{aligned}$$

Example (continued)

The last expression is written in terms of the standardized values $(x_1 - \mu_1)/\sqrt{\sigma_{11}}$ and $(x_2 - \mu_2)/\sqrt{\sigma_{22}}$.

Next, since $|\Sigma| = \sigma_{11}\sigma_{22} - \sigma_{12}^2 = \sigma_{11}\sigma_{22}(1 - \rho_{12}^2)$, we can substitute for Σ^{-1} and $|\Sigma|$ to get the expression for the bivariate normal density:

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{|\Sigma|}} \exp\left[-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu)\right].$$

Hence, in terms of the individual parameters,

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{\sigma_{11}\sigma_{22}(1 - \rho_{12}^2)}} \exp\left\{-\frac{1}{2(1 - \rho_{12}^2)} \left[\left(\frac{x_1 - \mu_1}{\sqrt{\sigma_{11}}}\right)^2 + \left(\frac{x_2 - \mu_2}{\sqrt{\sigma_{22}}}\right)^2 - 2\rho_{12} \left(\frac{x_1 - \mu_1}{\sqrt{\sigma_{11}}}\right) \left(\frac{x_2 - \mu_2}{\sqrt{\sigma_{22}}}\right) \right] \right\}.$$

Properties

The following are true for a random vector X having a multivariate normal distribution.

- Linear combinations of the components of X are normally distributed.
- All subsets of the components of X have a (multivariate) normal distribution.
- Zero covariance implies that the corresponding components are independently distributed.
- The conditional distributions of the components are (multivariate) normal.

Remark

Consider two random variables X and Y . If the marginal distribution of each is normal, then their joint distribution need **not** be a (multivariate) normal distribution.

Example

We illustrate with an example showing that *normally distributed and uncorrelated does not imply jointly normal or independent*.

$$V \sim \text{Bern}\left(\frac{1}{2}\right), \quad W = 2V - 1$$

$$X \sim \mathcal{N}(0, 1), \quad Y = WX$$

Example (continued)

Under these conditions, it can be shown that

$$X \sim \mathcal{N}(0, 1) \quad \text{and} \quad Y \sim \mathcal{N}(0, 1).$$

Now

$$\begin{aligned} F_Y(\alpha) &= P(Y < \alpha) \\ &= P(Y < \alpha, W = 1) + P(Y < \alpha, W = -1) \end{aligned}$$

But

$$\begin{aligned} P(Y < \alpha, W = 1) &= P(X < \alpha, W = 1) \\ &= P(X < \alpha \mid W = 1)P(W = 1) \\ &= P(X < \alpha) \times \frac{1}{2} \end{aligned}$$

Example (continued)

Similarly, as $X \sim \mathcal{N}(0, 1)$, it is clear that

$$P(Y < \alpha, W = -1) = \frac{1}{2} \times P(X > \alpha)$$

Therefore,

$$\begin{aligned} F_Y(\alpha) &= \frac{1}{2} \{P(X < \alpha) + P(X > \alpha)\} \\ &= F_X(\alpha) \end{aligned}$$

Theorem

If $X = (X_1, \dots, X_p)^\top$ is distributed as $\mathcal{N}_p(\mu, \Sigma)$ (where $\mu = \mu_{p \times 1}$ and $\Sigma = \Sigma_{p \times p}$), then any linear combination of variables $a^\top X = a_1 X_1 + a_2 X_2 + \dots + a_p X_p$ (where $a = [a_1, \dots, a_p]^\top$), is distributed as $\mathcal{N}(a^\top \mu, a^\top \Sigma a)$. Also, if $a^\top X$ is distributed as $\mathcal{N}(a^\top \mu, a^\top \Sigma a)$ for every a , then X must be $\mathcal{N}_p(\mu, \Sigma)$.

Theorem

If $X = (X_1, \dots, X_p)^\top$ is distributed as $\mathcal{N}_p(\mu, \Sigma)$, the q linear combinations

$$A_{(q \times p)} X_{(p \times 1)} = \begin{bmatrix} a_{11}X_1 + \dots + a_{1p}X_p \\ a_{21}X_1 + \dots + a_{2p}X_p \\ \vdots \\ a_{q1}X_1 + \dots + a_{qp}X_p \end{bmatrix}$$

are distributed as $\mathcal{N}_q(A\mu, A\Sigma A^\top)$.

Also, $X_{(p \times 1)} + d_{(p \times 1)}$, where d is a vector of constants, is distributed as $\mathcal{N}_p(\mu + d, \Sigma)$.

Theorem

All subsets of X are normally distributed. If we respectively partition X , its mean vector μ , and its covariance matrix Σ as

$$X_{(p \times 1)} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}, \quad \mu_{(p \times 1)} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix},$$

where X_1 and μ_1 are $(q \times 1)$ vectors and X_2 and μ_2 are $((p - q) \times 1)$ vectors, and

$$\Sigma_{(p \times p)} = \begin{bmatrix} \Sigma_{11(q \times q)} & \vdots & \Sigma_{12(q \times (p-q))} \\ \Sigma_{21(p-q) \times q} & \vdots & \Sigma_{22(p-q) \times (p-q)} \end{bmatrix},$$

Then X_1 is distributed as

$$X_1 \sim \mathcal{N}_q(\mu_1, \Sigma_{11}).$$

Theorem

- If $X_{1(q_1 \times 1)}$ and $X_{2(q_2 \times 1)}$ are independent if, then $\text{Cov}(X_1, X_2) = 0$, a $q_1 \times q_2$ matrix of zeros.
- If

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}_{q_1+q_2} \left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \Sigma_{11} & : & \Sigma_{12} \\ \Sigma_{21} & : & \Sigma_{22} \end{bmatrix} \right),$$

then X_1 and X_2 are independent if and only if $\Sigma_{12} = 0$.

Example

Let

$$X_{(3 \times 1)} \sim \mathcal{N}_3(\mu, \Sigma) \quad \text{with} \quad \Sigma = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Are X_1 and X_2 independent? What about (X_1, X_2) and X_3 ?

Answer

Since X_1 and X_2 have covariance $\sigma_{12} = 1$, they are **not independent**. However, partitioning X and Σ as

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix}, \quad \Sigma = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{bmatrix},$$

where

$$\Sigma_{11} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}, \quad \Sigma_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \Sigma_{22} = [2],$$

we see that $X_1 = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$ and X_3 have covariance matrix

$$\Sigma_{12} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Therefore, (X_1, X_2) and X_3 are independent. This implies X_3 is independent of X_1 and also of X_2 .

Theorem

Let X be distributed as $\mathcal{N}_p(\mu, \Sigma)$ with $|\Sigma| > 0$. Then:

- The quadratic form

$$(X - \mu)^\top \Sigma^{-1} (X - \mu)$$

is distributed as χ_p^2 , where χ_p^2 denotes the chi-square distribution with p degrees of freedom.

- The distribution $\mathcal{N}_p(\mu, \Sigma)$ assigns probability $1 - \alpha$ to the solid ellipsoid

$$\{\mathbf{x} : (\mathbf{x} - \mu)^\top \Sigma^{-1} (\mathbf{x} - \mu) \leq \chi_p^2(\alpha)\},$$

where $\chi_p^2(\alpha)$ denotes the upper (100α) th percentile of the χ_p^2 distribution.

Theorem

Let $X_1, X_2, \dots, X_n$ be mutually independent with X_j distributed as $\mathcal{N}_p(\mu_j, \Sigma)$. (Note that each X_j has the same covariance matrix Σ .) Then

$$V_1 = c_1 X_1 + c_2 X_2 + \dots + c_n X_n$$

is distributed as

$$\mathcal{N}_p \left(\sum_{j=1}^n c_j \mu_j, \left(\sum_{j=1}^n c_j^2 \right) \Sigma \right).$$

Moreover,

$$V_1 = c_1 X_1 + c_2 X_2 + \dots + c_n X_n \quad \text{and} \quad V_2 = b_1 X_1 + b_2 X_2 + \dots + b_n X_n$$

are jointly multivariate normal with covariance matrix

$$\begin{bmatrix} \left(\sum_{j=1}^n c_j^2 \right) \Sigma & (b^\top c) \Sigma \\ (b^\top c) \Sigma & \left(\sum_{j=1}^n b_j^2 \right) \Sigma \end{bmatrix}.$$

Consequently, V_1 and V_2 are independent if

$$b^\top c = \sum_{j=1}^n c_j b_j = 0.$$

Constant probability density contour

- Constant probability density contour = $\{ \text{all } x \text{ such that } \{(x - \mu)^\top \Sigma^{-1}(x - \mu) = c^2 \}.$
= surface of an ellipsoid centered at μ .
- The axes of each ellipsoid of constant density are in the direction of the eigenvectors of Σ^{-1} and their lengths are proportional to the reciprocals of the square roots of the eigenvalues of Σ^{-1} .

Example (Contours of the bivariate normal density)

We shall obtain the axes of constant probability density contours for a bivariate normal distribution when $\sigma_{11} = \sigma_{22}$. From $|\Sigma - \lambda I| = 0$, we have

$$0 = \begin{vmatrix} \sigma_{11} - \lambda & \sigma_{12} \\ \sigma_{12} & \sigma_{11} - \lambda \end{vmatrix} = (\sigma_{11} - \lambda)^2 - \sigma_{12}^2 = (\lambda - \sigma_{11} - \sigma_{12})(\lambda - \sigma_{11} + \sigma_{12}).$$

Consequently, the eigenvalues are $\lambda_1 = \sigma_{11} + \sigma_{12}$ and $\lambda_2 = \sigma_{11} - \sigma_{12}$.

The eigenvector e_1 is determined from

$$\begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{12} & \sigma_{11} \end{bmatrix} \begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = (\sigma_{11} + \sigma_{12}) \begin{bmatrix} e_1 \\ e_2 \end{bmatrix},$$

or

$$\sigma_{11}e_1 + \sigma_{12}e_2 = (\sigma_{11} + \sigma_{12})e_1, \quad \sigma_{12}e_1 + \sigma_{11}e_2 = (\sigma_{11} + \sigma_{12})e_2.$$

These imply $e_1 = e_2$, and after normalization, the first eigenvalue–eigenvector pair is

$$\lambda_1 = \sigma_{11} + \sigma_{12}, \quad e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}.$$

Example (Continued)

Similarly, $\lambda_2 = \sigma_{11} - \sigma_{12}$ yields the eigenvector

$$e_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}.$$

When the covariance σ_{12} (or correlation ρ_{12}) is positive, $\lambda_1 = \sigma_{11} + \sigma_{12}$ is the largest eigenvalue, and its associated eigenvector $e_1 = [1/\sqrt{2}, 1/\sqrt{2}]^\top$ lies along the 45° line through the point $\mu' = [\mu_1, \mu_2]^\top$. Hence, for positively correlated normal random variables, the major axis of the constant-density ellipses lies along the 45° line through μ .

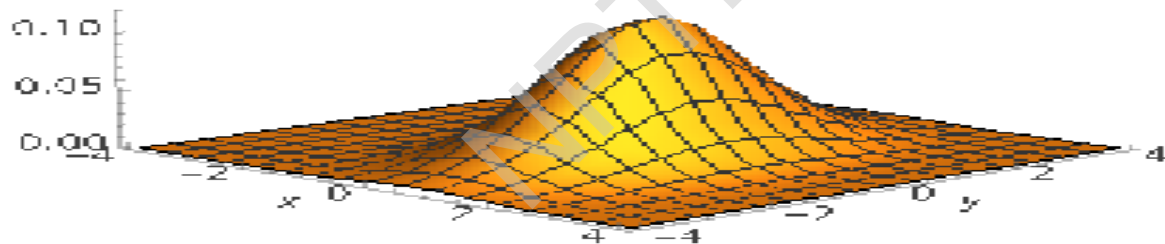
Example (Continued)

When the covariance (correlation) is negative, $\lambda_2 = \sigma_{11} - \sigma_{12}$ will be the largest eigenvalue, and the major axes of the constant-density ellipses will lie along a line at right angles to the 45° line through μ . (These results are true only for $\sigma_{11} = \sigma_{22}$.)

To summarize, the axes of the ellipses of constant density for a bivariate normal distribution with $\sigma_{11} = \sigma_{22}$ are determined by

$$\pm c\sqrt{\sigma_{11} + \sigma_{12}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \quad \text{and} \quad \pm c\sqrt{\sigma_{11} - \sigma_{12}} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}.$$

μ_X	μ_Y	σ_X	σ_Y	ρ
0.395	-0.115	0.878	1.516	-0.146



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Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

BY
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Module 07: Introduction to Multivariate Distribution

Lecture 04: Multivariate Normal
Distribution 2



Multivariate Data

Consider n i.i.d. sample $x_1, \dots, x_n$ from p - dimensional distribution where $y_i = [x_{1i}, \dots, x_{ni}]^\top$ for each $i = 1, \dots, p$.
Let

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1p} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2p} \\ x_{31} & x_{32} & x_{33} & \cdots & x_{3p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{np} \end{pmatrix}$$

and let $\mathbf{1} = [\mathbf{1}]_{n \times 1} = [1, \dots, 1]^\top$

Multivariate Data (continued)

$$\bar{X}_i = (x_{1i} \cdot 1 + x_{2i} \cdot 1 + \cdots + x_{ni} \cdot 1)/n = (y_i^\top \cdot \mathbf{1})/n$$

$$\bar{X} = \begin{bmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \vdots \\ \bar{X}_p \end{bmatrix} = \frac{1}{n} \begin{bmatrix} y_1^\top \cdot \mathbf{1} \\ y_2^\top \cdot \mathbf{1} \\ \vdots \\ y_p^\top \cdot \mathbf{1} \end{bmatrix} = \frac{1}{n} \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{p1} & x_{p2} & \cdots & x_{pn} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

or

$$\bar{X} = \frac{1}{n} X^\top \mathbf{1}$$

That is, $\bar{X}$ is calculated from the transposed data matrix by postmultiplying by vector $\mathbf{1}$ and then multiplying the result by the constant $1/n$.

Multivariate Data (continued)

Next, we create an $n \times p$ matrix :

$$\mathbf{1}\bar{X}^T = \frac{1}{n}\mathbf{1}\mathbf{1}^T X = \begin{bmatrix} \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \\ \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \\ \vdots & \vdots & \ddots & \vdots \\ \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \end{bmatrix}$$

Subtracting this result from X produces the $n \times p$ matrix of deviations (residuals):

$$X - \frac{1}{n}\mathbf{1}\mathbf{1}^T X = \begin{bmatrix} X_{11} - \bar{X}_1 & X_{12} - \bar{X}_2 & \cdots & X_{1p} - \bar{X}_p \\ X_{21} - \bar{X}_1 & X_{22} - \bar{X}_2 & \cdots & X_{2p} - \bar{X}_p \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} - \bar{X}_1 & X_{n2} - \bar{X}_2 & \cdots & X_{np} - \bar{X}_p \end{bmatrix}$$

Multivariate Data (continued)

Now, the matrix $(n-1)S$ representing sums of squares and cross products is:

$$\begin{aligned}(n-1)S &= \begin{bmatrix} x_{11} - \bar{X}_1 & x_{21} - \bar{X}_1 & \cdots & x_{n1} - \bar{X}_1 \\ x_{12} - \bar{X}_2 & x_{22} - \bar{X}_2 & \cdots & x_{n2} - \bar{X}_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_{1p} - \bar{X}_p & x_{2p} - \bar{X}_p & \cdots & x_{np} - \bar{X}_p \end{bmatrix} \begin{bmatrix} x_{11} - \bar{X}_1 & x_{12} - \bar{X}_2 & \cdots & x_{1p} - \bar{X}_p \\ x_{21} - \bar{X}_1 & x_{22} - \bar{X}_2 & \cdots & x_{2p} - \bar{X}_p \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} - \bar{X}_1 & x_{n2} - \bar{X}_2 & \cdots & x_{np} - \bar{X}_p \end{bmatrix} \\ &= \left(X - \frac{1}{n} \mathbf{1}\mathbf{1}^\top X \right)^\top \left(X - \frac{1}{n} \mathbf{1}\mathbf{1}^\top X \right) = X^\top \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) X\end{aligned}$$

since

$$\left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) \left(I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top - \frac{1}{n} \mathbf{1}\mathbf{1}^\top + \frac{1}{n^2} \mathbf{1}\mathbf{1}^\top \mathbf{1}\mathbf{1}^\top = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$$

where I is the identity matrix of order n .

Here

$$S = \begin{pmatrix} s_{11} & s_{12} & s_{13} & \cdots & s_{1p} \\ s_{21} & s_{22} & s_{23} & \cdots & s_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{n1} & s_{n2} & s_{n3} & \cdots & s_{np} \end{pmatrix} \quad \{s_{ik} = \frac{1}{n-1} \sum_{j=1}^n (x_{ji} - \bar{X}_i)(x_{jk} - \bar{X}_i)\}$$

Multivariate Data (continued)

To summarize, the matrix expressions relating $\bar{X}$ and S to the data set X are

$$\bar{X} = \frac{1}{n} X^T \cdot \mathbf{1}$$

$$S = \frac{1}{n-1} X^T \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) X$$

Theorem

Let $X_1, X_2, \dots, X_n$ be a random sample from a joint distribution with mean vector μ and covariance matrix Σ . Then $\bar{X}$ is an unbiased estimator of μ , and its covariance matrix is $\frac{1}{n}\Sigma$.

That is,

$$E(\bar{X}) = \mu \quad (\text{population mean vector})$$

$$\text{Cov}(\bar{X}) = \frac{1}{n}\Sigma \quad (\text{population variance-covariance matrix divided by sample size})$$

For the covariance matrix S_n

$$\left(\text{where } S_n = \frac{n-1}{n}S \right)$$

$$E(S_n) = \frac{n-1}{n}\Sigma = \Sigma - \frac{1}{n}\Sigma$$

Theorem (continued)

Thus,

$$E\left(\frac{n}{n-1} S_n\right) = \Sigma$$

so $\left[\frac{n}{(n-1)}\right] S_n$ is an unbiased estimator of Σ , while S_n is a biased estimator with

$$\text{bias} = E(S_n) - \Sigma = -\frac{1}{n}\Sigma.$$

Proof

Since $\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$, we have

$$E(\bar{X}) = E\left(\frac{1}{n}X_1 + \frac{1}{n}X_2 + \cdots + \frac{1}{n}X_n\right) = \frac{1}{n}E(X_1) + \frac{1}{n}E(X_2) + \cdots + \frac{1}{n}E(X_n) = \mu$$

Next,

$$\begin{aligned}(\bar{X} - \mu)(\bar{X} - \mu)^\top &= \left(\frac{1}{n} \sum_{j=1}^n (X_j - \mu)\right) \left(\frac{1}{n} \sum_{\ell=1}^n (X_\ell - \mu)\right)^\top \\ &= \frac{1}{n^2} \sum_{j=1}^n \sum_{\ell=1}^n (X_j - \mu)(X_\ell - \mu)^\top.\end{aligned}$$

Proof (continued)

$$\text{Cov}(\bar{X}) = E(\bar{X} - \mu)(\bar{X} - \mu)^\top = \frac{1}{n^2} E \left(\sum_{j=1}^n (X_j - \mu) \right) \left(\sum_{\ell=1}^n (X_\ell - \mu) \right)^\top$$

For $j \neq \ell$, each entry in $E(X_j - \mu)(X_\ell - \mu)^\top$ is zero because the entry is the covariance between a component of X_j and a component of X_ℓ , and these are independent. Therefore,

$$\text{Cov}(\bar{X}) = \frac{1}{n^2} \sum_{j=1}^n E[(X_j - \mu)(X_j - \mu)^\top]$$

Since $\Sigma = E(X_j - \mu)(X_j - \mu)^\top$ is the common population covariance matrix for each X_j , we have

$$\text{Cov}(\bar{X}) = \frac{1}{n^2} \sum_{j=1}^n \Sigma = \frac{1}{n^2} (n\Sigma) = \left(\frac{1}{n} \right) \Sigma$$

Proof (continued)

To obtain the expected value of S_n , we first note that $(X_j - \bar{X})(X_j - \bar{X})^\top$ is the (i, k) -th element of $(X_j - \bar{X})(X_j - \bar{X})^\top$. The matrix representing sums of squares and cross products can be written as

$$\begin{aligned}\sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top &= \sum_{j=1}^n (X_j - \bar{X})X_j^\top + \sum_{j=1}^n (X_j - \bar{X})(-\bar{X})^\top \\ &= \sum_{j=1}^n X_j X_j^\top - n\bar{X}\bar{X}^\top\end{aligned}$$

since $\sum_{j=1}^n (X_j - \bar{X}) = 0$ and $n\bar{X} = \sum_{j=1}^n X_j$.

Proof (continued)

Therefore, its expected value is

$$E \left(\sum_{j=1}^n X_j X_j^\top - n \bar{X} \bar{X}^\top \right) = \sum_{j=1}^n E(X_j X_j^\top) - n E(\bar{X} \bar{X}^\top)$$

For any random vector V with $E(V) = \mu_V$ and $\text{Cov}(V) = \Sigma_V$, we have $E(VV^\top) = \Sigma_V + \mu_V \mu_V^\top$. Applying this, we obtain

$$E(X_j X_j^\top) = \Sigma + \mu \mu^\top \quad \text{and} \quad E(\bar{X} \bar{X}^\top) = \frac{1}{n} \Sigma + \mu \mu^\top$$

Proof (continued)

Using these results, we get

$$\sum_{j=1}^n E(X_j X_j^\top) - nE(\bar{X} \bar{X}^\top) = n\Sigma + n\mu\mu^\top - n\left(\frac{1}{n}\Sigma + \mu\mu^\top\right) = (n-1)\Sigma$$

and thus, since $S_n = \frac{1}{n} \left(\sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top \right)$, it follows immediately

$$E(S_n) = \frac{n-1}{n}\Sigma$$

Central Limit Theorem

Let $Y_1, \dots, Y_n$ be independent observations from a population with mean $E(Y_i) = \mu$, and finite (nonsingular) covariance Σ . Then

$\sqrt{n}(\bar{X} - \mu)$ is approximate $\mathcal{N}_p(0, \Sigma)$
and

$n(\bar{X} - \mu)^\top S^{-1}(\bar{X} - \mu)$ is approximately χ_p^2 for $n - p$ large.

Maximum Likelihood Estimation For Normal Distribution

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ and covariance Σ . Then

$$\hat{\mu} = \bar{X} \quad \text{and} \quad \hat{\Sigma} = \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top = \frac{(n-1)}{n} S$$

are the maximum likelihood estimators of μ and Σ , respectively. Their observed values, $\bar{x}$ and $(1/n) \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top$, are called the maximum likelihood estimates of μ and Σ .

The Sampling of $\bar{X}$ and S

- The tentative assumption that $X_1, X_2, \dots, X_n$ constitute a random sample from a normal population with mean μ and covariance Σ completely determines the sampling distributions of $\bar{X}$ and S .
- In the univariate case, we know that $\bar{X}$ is normal with the mean μ and covariance $\frac{1}{n}\Sigma$.
- For the sample variance, recall that $s^2 = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})^2$ is distributed as σ^2 times a chi-square variable having $n - 1$ degrees of freedom.

The Sampling Distribution of $\bar{X}$ and S (continued)

Wishart Distribution: The sampling distribution of the sample covariance matrix is called the *Wishart distribution*, it is defined as the sum of independent products of multivariate normal random vectors.

Wishart distribution with m d.f. = distribution of $\sum_{i=1}^m Z_i Z_i^\top$

where the Z_i are each independently distributed as $\mathcal{N}_p(0, \Sigma)$.

The Sampling of $\bar{X}$ and S (continued)

We summarize the sampling distribution results as follows:

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a p -variate normal distribution with mean μ and covariance matrix Σ .
Then

- 1 $\bar{X}$ is distributed as $\mathcal{N}_p(\mu, (1/n)\Sigma)$.
- 2 $(n-1)S$ is distributed as a Wishart random matrix with $n-1$ degrees of freedom.
- 3 $\bar{X}$ and S are independent.

Because Σ is unknown, the distribution of $\bar{X}$ cannot be used directly to make inferences about μ . However, S provides independent information about Σ , and the distribution of S does not depend on μ . This allows us to construct a statistic for making inferences about μ .

Hypothesis Testing

Let us start by recalling the univariate theory for hypothesis testing for μ at μ_0 , where the null hypothesis is $H_0 : \mu = \mu_0$ and the alternate hypothesis is $H_1 : \mu \neq \mu_0$. In general, the correspondence between acceptance regions for the tests of $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ and the confidence interval of μ ,

$$\left\{ \text{Do not reject } H_0 : \mu = \mu_0 \text{ at level } \alpha \right\} \quad \text{or} \quad \left| \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \right| \leq t_{n-1}(\alpha/2)$$

is equivalent to

$$\left\{ \mu_0 \text{ lies in the } 100(1 - \alpha)\% \text{ confidence interval } \bar{x} \pm t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}} \right\}$$

or, equivalently,

$$\bar{x} - t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}} \leq \mu_0 \leq \bar{x} + t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}}.$$

Hypothesis Testing (continued)

Let $X_1, X_2, \dots, X_n$ be a random sample from an $\mathcal{N}_p(\mu, \Sigma)$ population. Then

$$\bar{X} = \frac{1}{n} \sum_{j=1}^n X_j,$$

$$S = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top.$$

$$\alpha = P\left[T^2 > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)\right] = P\left[n(\bar{X} - \mu)^\top S^{-1}(\bar{X} - \mu) > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)\right].$$

Here $F_{p, n-p}(\alpha)$ is the upper $100\alpha\%$ percentile of the $F_{p, n-p}$ distribution.

This leads to the test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$.

Hypothesis Testing (continued)

Let $T^2 = (\bar{X} - \mu_0)^\top \left(\frac{1}{n}S\right) (\bar{X} - \mu_0) = n(\bar{X} - \mu_0)^\top S^{-1}(\bar{X} - \mu_0)$,

where $\bar{X}_{p \times 1} = \frac{1}{n} \sum_{j=1}^n X_j$, $S_{p \times p} = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top$ and $\mu_{0p \times 1} = [\mu_{10}, \dots, \mu_{p0}]^\top$

At the level of significance α level of significance, we reject H_0 in favor of H_1 if the observed

$$T^2 = n(\bar{X} - \mu_0)^\top S^{-1}(\bar{X} - \mu_0) > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$$

where the statistic T^2 is called *Hotelling's T^2* distribution (it follows $\frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$ distribution).

QQ plot

Special plots called Q-Q plots can be used to assess the assumption of normality. It is constructed as

- Sort the sample data to obtain empirical quantiles.
- Compute corresponding quantiles of a reference distribution.
- Plot empirical quantiles on the vertical axis and theoretical on the horizontal axis.

To interpret this

- Points near a straight line imply a good match to the reference distribution.
- Curvature indicates skewness or shape mismatch.
- Points deviating in tails indicate heavy or light tails.

QQ plot (continued)

To simplify this

- Let $x_1, x_2, \dots, x_n$ represent n observations on a characteristic X_i .
- Let $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ denote these observations ordered by magnitude.
- The values $x_{(j)}$ are called sample quantiles, where $x_{(1)}$ is the smallest and $x_{(n)}$ is the largest.
- When all $x_{(j)}$ are distinct, exactly j observations are less than or equal to $x_{(j)}$.
- The proportion of observations less than or equal to $x_{(j)}$ is approximately $(j - \frac{1}{2}) / n$.
- For a standard normal distribution, quantiles $q(j)$ satisfy

$$P[Z \leq q(j)] = \int_{-\infty}^{q(j)} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = p(j) = \frac{j - \frac{1}{2}}{n}.$$

- Here $p(j)$ is the probability of obtaining a value less than or equal to $q(j)$ from a standard normal draw.
- The idea of the QQ diagram is to look at pairs $(q(j), x_{(j)})$ with the same cumulative probability $(j - \frac{1}{2}) / n$.
- If the data follow a normal population, the points $(q(j), x_{(j)})$ lie approximately on a straight line.
- This occurs because $\sigma q(j) + \mu$ is nearly the expected sample quantile.

Example

A sample of $n = 10$ observations is given. The ordered observations $x_{(j)}$, the probability levels $(j - \frac{1}{2})/n$, and the corresponding standard normal quantiles $q(j)$ are shown below.

Ordered observations $x_{(j)}$	$(j - \frac{1}{2})/n$	Standard normal quantiles $q(j)$
-1.00	0.05	-1.645
-0.10	0.15	-1.036
0.16	0.25	-0.674
0.41	0.35	-0.385
0.62	0.45	-0.125
0.85	0.55	0.125
1.26	0.65	0.385
1.54	0.75	0.674
1.71	0.85	1.036
2.30	0.95	1.645

Example (continued)

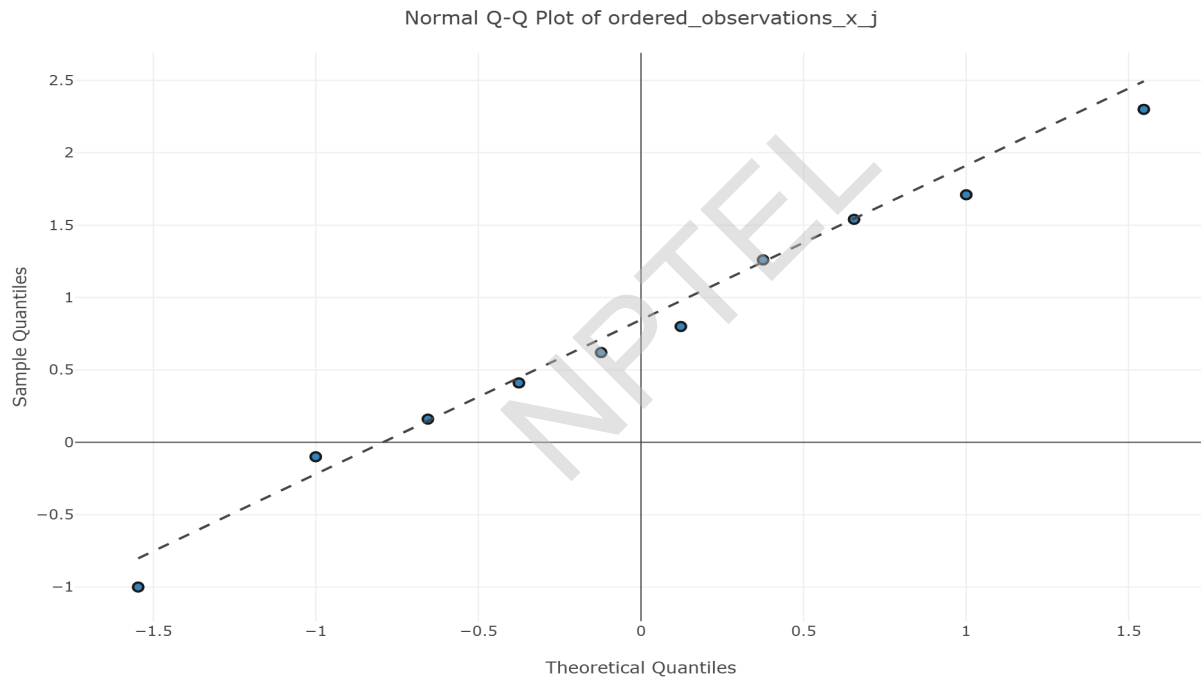
For example,

$$P[Z \leq 0.385] = \int_{-\infty}^{0.385} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 0.65.$$

This matches the probability level $(j - \frac{1}{2})/n$.

The Q-Q plot constructed from these values plots $x_{(j)}$ against $q(j)$. The points lie close to a straight line, indicating that the sample is consistent with a normal population, even with $n = 10$.

Plot



Maximum Likelihood Estimate for Multivariate Normal Distribution

Let us assume that the $p \times 1$ vectors $X_1, X_2, \dots, X_n$ represent a random sample from a multivariate normal population with mean vector μ and covariance matrix Σ . Since $X_1, X_2, \dots, X_n$ are mutually independent and each has distribution $\mathcal{N}_p(\mu, \Sigma)$, the joint density function of all the observations is the product of the marginal normal densities:

$$\begin{aligned}\{\text{Joint density of } X_1, X_2, \dots, X_n\} &= \prod_{j=1}^n \left[\frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} e^{-(x_j - \mu)^\top \Sigma^{-1} (x_j - \mu)/2} \right] \\ &= \frac{1}{(2\pi)^{np/2} |\Sigma|^{n/2}} e^{-\frac{1}{2} \sum_{j=1}^n (x_j - \mu)^\top \Sigma^{-1} (x_j - \mu)}\end{aligned}$$

When the numerical values of the observations become available, they may be substituted for the x_j in the above equation. The resulting expression, now considered as a function of μ and Σ for the fixed set of observations $x_1, x_2, \dots, x_n$, is called the likelihood.

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

Now the exponent in the joint density can be simplified.

$$(x_j - \mu)^\top \Sigma^{-1} (x_j - \mu) = \text{tr}[(x_j - \mu)^\top \Sigma^{-1} (x_j - \mu)] = \text{tr}[\Sigma^{-1} (x_j - \mu)(x_j - \mu)^\top]$$

Next,

$$\sum_{j=1}^n (x_j - \mu)^\top \Sigma^{-1} (x_j - \mu) = \sum_{j=1}^n \text{tr}[(x_j - \mu)^\top \Sigma^{-1} (x_j - \mu)] = \sum_{j=1}^n \text{tr}[\Sigma^{-1} (x_j - \mu)(x_j - \mu)^\top] = \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \mu)(x_j - \mu)^\top \right) \right]$$

Since the trace of a sum of matrices is equal to the sum of the traces of the matrices.

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

We can add and subtract $\bar{x} = (1/n) \sum_{j=1}^n x_j$ in each term $(x_j - \mu)$ in $\sum_{j=1}^n (x_j - \mu)(x_j - \mu)^\top$ to give

$$\sum_{j=1}^n (x_j - \bar{x} + \bar{x} - \mu)(x_j - \bar{x} + \bar{x} - \mu)^\top = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + \sum_{j=1}^n (\bar{x} - \mu)(\bar{x} - \mu)^\top = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + n(\bar{x} - \mu)(\bar{x} - \mu)^\top$$

because the cross-product terms, $[\sum_{j=1}^n (x_j - \bar{x})(\bar{x} - \mu)^\top$ and $\sum_{j=1}^n (\bar{x} - \mu)(x_j - \bar{x})^\top$, are both matrices of zeros.

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

Consequently, using Equations above, we can write the joint density of a random sample from a multivariate normal population as

$$\begin{aligned} & \{ \text{Joint density of } X_1, X_2, \dots, X_n \} \\ &= (2\pi)^{-np/2} |\Sigma|^{-n/2} \\ & \times \exp \left\{ -\text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + n(\bar{x} - \mu)(\bar{x} - \mu)^\top \right) \right] / 2 \right\} \end{aligned}$$

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

Substituting the observed values $x_1, x_2, \dots, x_n$ into the joint density yields the likelihood function. We shall denote this function by $L(\mu, \Sigma)$, to stress the fact that it is a function of the (unknown) population parameters μ and Σ . Thus, when the vectors x_j contain the specific numbers actually observed, we have

$$L(\mu, \Sigma) = \frac{1}{(2\pi)^{np/2} |\Sigma|^{n/2}} \exp \left\{ -\frac{1}{2} \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + n(\bar{x} - \mu)(\bar{x} - \mu)^\top \right) \right] \right\}$$

It will be convenient to use the exponent in the likelihood function in different ways. In particular, we shall make use of the identity

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

$$\begin{aligned} & \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + n(\bar{x} - \mu)(\bar{x} - \mu)^\top \right) \right] \\ &= \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top \right) \right] + n \text{tr} [\Sigma^{-1}(\bar{x} - \mu)(\bar{x} - \mu)^\top] \\ &= \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top \right) \right] + n(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu) \end{aligned}$$

Theorem

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ and covariance Σ . Then

$$\hat{\mu} = \bar{X} \quad \text{and} \quad \hat{\Sigma} = \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top = \frac{(n-1)}{n} S$$

are the maximum likelihood estimators of μ and Σ , respectively. Their observed values, $\bar{x}$ and $(1/n) \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top$, are called the maximum likelihood estimates of μ and Σ .

Theorem (Proof)

Proof. The exponent in the likelihood function, apart from the multiplicative factor $-\frac{1}{2}$, is

$$\text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top \right) \right] + n(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu)$$

Σ^{-1} is positive definite, so the distance $(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu) > 0$ unless $\mu = \bar{x}$. Thus, the likelihood is maximized with respect to μ at $\hat{\mu} = \bar{x}$. It remains to maximize

$$L(\mu, \Sigma) = \frac{1}{(2\pi)^{np/2} |\Sigma|^{n/2}} \exp \left\{ -\frac{1}{2} \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top \right) \right] \right\}$$

over Σ .

Theorem (Proof)

By the following result:

Given a $p \times p$ symmetric positive definite matrix B and a scalar $b > 0$,

$$\frac{1}{|\Sigma|^b} e^{-\text{tr}(\Sigma^{-1}B)/2} \leq \frac{1}{B^b} (2b)^{pb} e^{-bp}$$

for all positive definite $\Sigma_{p \times p}$, with equality holding only for $\Sigma = (1/2b)B$.

with $b = n/2$ and

$$B = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top,$$

the maximum occurs at

$$\hat{\Sigma} = \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top.$$

The maximum likelihood estimators are random quantities. They are obtained by replacing the observations $x_1, x_2, \dots, x_n$ in the expressions for $\hat{\mu}$ and $\hat{\Sigma}$ with the corresponding random vectors $X_1, X_2, \dots, X_n$.

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Thank you





SWAYAM NPTEL COURSE ON STATISTICAL FOUNDATION FOR BIG DATA ANALYSIS

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Module 07: Introduction to Multivariate Distribution

Lecture 05: Multivariate Normal
Distribution 3



Multivariate Data

Consider n i.i.d. sample $x_1, \dots, x_n$ from p - dimensional distribution where $y_i = [x_{1i}, \dots, x_{pi}]^\top$ for each $i = 1, \dots, p$.
Let

$$X = \begin{pmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1p} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2p} \\ x_{31} & x_{32} & x_{33} & \cdots & x_{3p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{np} \end{pmatrix}$$

and let $\mathbf{1} = [\mathbf{1}]_{n \times 1} = [1, \dots, 1]^\top$

Multivariate Data (continued)

$$\bar{X}_i = (x_{1i} \cdot 1 + x_{2i} \cdot 1 + \cdots + x_{ni} \cdot 1)/n = (y_i^\top \cdot \mathbf{1})/n$$

$$\bar{X} = \begin{bmatrix} \bar{X}_1 \\ \bar{X}_2 \\ \vdots \\ \bar{X}_p \end{bmatrix} = \frac{1}{n} \begin{bmatrix} y_1^\top \cdot \mathbf{1} \\ y_2^\top \cdot \mathbf{1} \\ \vdots \\ y_p^\top \cdot \mathbf{1} \end{bmatrix} = \frac{1}{n} \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{p1} & x_{p2} & \cdots & x_{pn} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$$

or

$$\bar{X} = \frac{1}{n} X^\top \mathbf{1}$$

That is, $\bar{X}$ is calculated from the transposed data matrix by postmultiplying by vector $\mathbf{1}$ and then multiplying the result by the constant $1/n$.

Multivariate Data (continued)

Next, we create an $n \times p$ matrix :

$$\mathbf{1}\bar{X}^T = \frac{1}{n}\mathbf{1}\mathbf{1}^T X = \begin{bmatrix} \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \\ \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \\ \vdots & \vdots & \ddots & \vdots \\ \bar{X}_1 & \bar{X}_2 & \cdots & \bar{X}_p \end{bmatrix}$$

Subtracting this result from X produces the $n \times p$ matrix of deviations (residuals):

$$X - \frac{1}{n}\mathbf{1}\mathbf{1}^T X = \begin{bmatrix} X_{11} - \bar{X}_1 & X_{12} - \bar{X}_2 & \cdots & X_{1p} - \bar{X}_p \\ X_{21} - \bar{X}_1 & X_{22} - \bar{X}_2 & \cdots & X_{2p} - \bar{X}_p \\ \vdots & \vdots & \ddots & \vdots \\ X_{n1} - \bar{X}_1 & X_{n2} - \bar{X}_2 & \cdots & X_{np} - \bar{X}_p \end{bmatrix}$$

Multivariate Data (continued)

Now, the matrix $(n-1)S$ representing sums of squares and cross products is:

$$\begin{aligned}(n-1)S &= \begin{bmatrix} x_{11} - \bar{X}_1 & x_{21} - \bar{X}_1 & \cdots & x_{n1} - \bar{X}_1 \\ x_{12} - \bar{X}_2 & x_{22} - \bar{X}_2 & \cdots & x_{n2} - \bar{X}_2 \\ \vdots & \vdots & \ddots & \vdots \\ x_{1p} - \bar{X}_p & x_{2p} - \bar{X}_p & \cdots & x_{np} - \bar{X}_p \end{bmatrix}^T \begin{bmatrix} x_{11} - \bar{X}_1 & x_{12} - \bar{X}_2 & \cdots & x_{1p} - \bar{X}_p \\ x_{21} - \bar{X}_1 & x_{22} - \bar{X}_2 & \cdots & x_{2p} - \bar{X}_p \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} - \bar{X}_1 & x_{n2} - \bar{X}_2 & \cdots & x_{np} - \bar{X}_p \end{bmatrix} \\ &= \left(X - \frac{1}{n} \mathbf{1} \mathbf{1}^T X \right)^T \left(X - \frac{1}{n} \mathbf{1} \mathbf{1}^T X \right) = X^T \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) X\end{aligned}$$

since

$$\left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) = I - \frac{1}{n} \mathbf{1} \mathbf{1}^T - \frac{1}{n} \mathbf{1} \mathbf{1}^T + \frac{1}{n^2} \mathbf{1} \mathbf{1}^T \mathbf{1} \mathbf{1}^T = I - \frac{1}{n} \mathbf{1} \mathbf{1}^T$$

where I is the identity matrix of order n .

Here

$$\begin{pmatrix} s_{11} & s_{12} & s_{13} & \cdots & s_{1p} \\ s_{12} & s_{22} & s_{23} & \cdots & s_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{1p} & s_{2p} & s_{3p} & \cdots & s_{pp} \end{pmatrix} \left\{ s_{ik} = \frac{1}{n-1} \sum_{j=1}^n (x_{ji} - \bar{X}_i)(x_{jk} - \bar{X}_k) \right\}$$

Multivariate Data (continued)

To summarize, the matrix expressions relating $\bar{X}$ and S to the data set X are

$$\bar{X} = \frac{1}{n} X^T \cdot \mathbf{1}$$

$$S = \frac{1}{n-1} X^T \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^T \right) X$$

Theorem

Let $X_1, X_2, \dots, X_n$ be a random sample from a joint distribution with mean vector $\mu_{p \times 1}$ and covariance matrix $\Sigma_{p \times p}$. Then $\bar{X}$ is an unbiased estimator of μ , and its covariance matrix is $\frac{1}{n}\Sigma$.

That is,

$$E(\bar{X}) = \mu \quad (\text{population mean vector})$$

$$\text{Cov}(\bar{X}) = \frac{1}{n}\Sigma \quad (\text{population variance-covariance matrix divided by sample size}).$$

If,

$$S = \begin{pmatrix} s_{11} & s_{12} & s_{13} & \cdots & s_{1p} \\ s_{12} & s_{22} & s_{23} & \cdots & s_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{1p} & s_{2p} & s_{3p} & \cdots & s_{pp} \end{pmatrix} \quad \{s_{ik} = \frac{1}{n-1} \sum_{j=1}^n (x_{ji} - \bar{X}_i)(x_{jk} - \bar{X}_i)\},$$

then $E(S) = \Sigma$

Proof

Since $\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$, we have

$$E(\bar{X}) = E\left(\frac{1}{n}X_1 + \frac{1}{n}X_2 + \cdots + \frac{1}{n}X_n\right) = \frac{1}{n}E(X_1) + \frac{1}{n}E(X_2) + \cdots + \frac{1}{n}E(X_n) = \mu$$

Next,

$$\begin{aligned}(\bar{X} - \mu)(\bar{X} - \mu)^\top &= \left(\frac{1}{n} \sum_{j=1}^n (X_j - \mu)\right) \left(\frac{1}{n} \sum_{\ell=1}^n (X_\ell - \mu)\right)^\top \\ &= \frac{1}{n^2} \sum_{j=1}^n \sum_{\ell=1}^n (X_j - \mu)(X_\ell - \mu)^\top.\end{aligned}$$

Proof (continued)

$$\text{Cov}(\bar{X}) = E(\bar{X} - \mu)(\bar{X} - \mu)^\top = \frac{1}{n^2} E \left(\sum_{j=1}^n (X_j - \mu) \right) \left(\sum_{\ell=1}^n (X_\ell - \mu) \right)^\top$$

For $j \neq \ell$, each entry in $E(X_j - \mu)(X_\ell - \mu)^\top$ is zero because the entry is the covariance between a component of X_j and a component of X_ℓ , and these are independent. Therefore,

$$\text{Cov}(\bar{X}) = \frac{1}{n^2} \sum_{j=1}^n E[(X_j - \mu)(X_j - \mu)^\top]$$

Since $\Sigma = E(X_j - \mu)(X_j - \mu)^\top$ is the common population covariance matrix for each X_j , we have

$$\text{Cov}(\bar{X}) = \frac{1}{n^2} \sum_{j=1}^n \Sigma = \frac{1}{n^2} (n\Sigma) = \left(\frac{1}{n} \right) \Sigma$$

Proof (continued)

To obtain the expected value of S_n , we first note that $(X_j - \bar{X})(X_j - \bar{X})^\top$ is the (i, k) -th element of $(X_j - \bar{X})(X_j - \bar{X})^\top$. The matrix representing sums of squares and cross products can be written as

$$\begin{aligned}\sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top &= \sum_{j=1}^n (X_j - \bar{X})X_j^\top + \sum_{j=1}^n (X_j - \bar{X})(-\bar{X})^\top \\ &= \sum_{j=1}^n X_j X_j^\top - n\bar{X}\bar{X}^\top\end{aligned}$$

since $\sum_{j=1}^n (X_j - \bar{X}) = 0$ and $n\bar{X} = \sum_{j=1}^n X_j$.

Proof (continued)

Therefore, its expected value is

$$E \left(\sum_{j=1}^n X_j X_j^\top - n \bar{X} \bar{X}^\top \right) = \sum_{j=1}^n E(X_j X_j^\top) - n E(\bar{X} \bar{X}^\top)$$

For any random vector V with $E(V) = \mu_V$ and $\text{Cov}(V) = \Sigma_V$, we have $E(VV^\top) = \Sigma_V + \mu_V \mu_V^\top$. Applying this, we obtain

$$E(X_j X_j^\top) = \Sigma + \mu \mu^\top \quad \text{and} \quad E(\bar{X} \bar{X}^\top) = \frac{1}{n} \Sigma + \mu \mu^\top$$

Proof (continued)

Using these results, we get

$$\sum_{j=1}^n E(X_j X_j^\top) - nE(\bar{X} \bar{X}^\top) = n\Sigma + n\mu\mu^\top - n\left(\frac{1}{n}\Sigma + \mu\mu^\top\right) = (n-1)\Sigma$$

and thus, since $S_n = \frac{1}{n} \left(\sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top \right)$, it follows immediately

$$E(S_n) = \frac{n-1}{n}\Sigma$$

Central Limit Theorem

Let $Y_1, \dots, Y_n$ be independent observations from a population with mean $\mu_{p \times 1}$, and finite (nonsingular) covariance matrix $\Sigma_{p \times p}$. Then $\sqrt{n}(\bar{Y} - \mu)$ is approximately $\mathcal{N}_p(0, \Sigma)$ when $n \rightarrow \infty$.

Maximum Likelihood Estimation For Normal Distribution

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ and covariance Σ . Then

$$\hat{\mu} = \bar{X} \quad \text{and} \quad \hat{\Sigma} = \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top = \frac{(n-1)}{n} S$$

are the maximum likelihood estimators of μ and Σ , respectively. Their observed values, $\bar{x}$ and $(1/n) \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top$, are called the maximum likelihood estimates of μ and Σ .

The Sampling of $\bar{X}$ and S

- The tentative assumption that $X_1, X_2, \dots, X_n$ constitute a random sample from a normal population with mean μ and covariance Σ completely determines the sampling distributions of $\bar{X}$ and S .
- In the univariate case, we know that $\bar{X}$ is normal with the mean μ and covariance $\frac{1}{n}\Sigma$.
- For the sample variance, recall that $s^2 = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})^2$ is distributed as σ^2 times a chi-square variable having $n - 1$ degrees of freedom.

The Sampling Distribution of $\bar{X}$ and S (continued)

Wishart Distribution: The sampling distribution of the sample covariance matrix is called the *Wishart distribution*, it is defined as the sum of independent products of multivariate normal random vectors.

Wishart distribution with m d.f. = distribution of $\sum_{i=1}^m Z_i Z_i^\top$

where the Z_i are each independently distributed as $\mathcal{N}_p(0, \Sigma)$.

The Sampling of $\bar{X}$ and S (continued)

We summarize the sampling distribution results as follows:

Let $X_1, X_2, \dots, X_n$ be a random sample of size n from a p -variate normal distribution with mean μ and covariance matrix Σ .
Then

- 1 $\bar{X}$ is distributed as $\mathcal{N}_p(\mu, (1/n)\Sigma)$.
- 2 $(n-1)S$ is distributed as a Wishart random matrix with $n-1$ degrees of freedom.
- 3 $\bar{X}$ and S are independent.

Because Σ is unknown, the distribution of $\bar{X}$ cannot be used directly to make inferences about μ . However, S provides independent information about Σ , and the distribution of S does not depend on μ . This allows us to construct a statistic for making inferences about μ .

Hypothesis Testing

Let us start by recalling the univariate theory for hypothesis testing for μ at μ_0 , where the null hypothesis is $H_0 : \mu = \mu_0$ and the alternate hypothesis is $H_1 : \mu \neq \mu_0$. In general, the correspondence between acceptance regions for the tests of $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ and the confidence interval of μ ,

$$\left\{ \text{Do not reject } H_0 : \mu = \mu_0 \text{ at level } \alpha \right\} \quad \text{or} \quad \left| \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \right| \leq t_{n-1}(\alpha/2)$$

is equivalent to

$$\left\{ \mu_0 \text{ lies in the } 100(1 - \alpha)\% \text{ confidence interval } \bar{x} \pm t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}} \right\}$$

or, equivalently,

$$\bar{x} - t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}} \leq \mu_0 \leq \bar{x} + t_{n-1}(\alpha/2) \frac{s}{\sqrt{n}}.$$

Hypothesis Testing (continued)

Let $X_1, X_2, \dots, X_n$ be a random sample from an $\mathcal{N}_p(\mu, \Sigma)$ population. Then

$$\bar{X} = \frac{1}{n} \sum_{j=1}^n X_j,$$

$$S = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top.$$

$$\alpha = P\left[T^2 > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)\right] = P\left[n(\bar{X} - \mu)^\top S^{-1}(\bar{X} - \mu) > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)\right].$$

Here $F_{p, n-p}(\alpha)$ is the upper $100\alpha\%$ percentile of the $F_{p, n-p}$ distribution.

This leads to the test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$.

Hypothesis Testing (continued)

Let $T^2 = (\bar{X} - \mu_0)^\top \left(\frac{1}{n}S\right) (\bar{X} - \mu_0) = n(\bar{X} - \mu_0)^\top S^{-1}(\bar{X} - \mu_0)$,

where $\bar{X}_{p \times 1} = \frac{1}{n} \sum_{j=1}^n X_j$, $S_{p \times p} = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top$ and $\mu_{0p \times 1} = [\mu_{10}, \dots, \mu_{p0}]^\top$

At the level of significance α level of significance, we reject H_0 in favor of H_1 if the observed

$$T^2 = n(\bar{X} - \mu_0)^\top S^{-1}(\bar{X} - \mu_0) > \frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$$

where the statistic T^2 is called *Hotelling's T^2* distribution (it follows $\frac{(n-1)p}{n-p} F_{p, n-p}(\alpha)$ distribution).

QQ plot

Special plots called Q-Q plots can be used to assess the assumption of normality. It is constructed as

- Sort the sample data to obtain empirical quantiles.
- Compute corresponding quantiles of a reference distribution.
- Plot empirical quantiles on the vertical axis and theoretical on the horizontal axis.

To interpret this

- Points near a straight line imply a good match to the reference distribution.
- Curvature indicates skewness or shape mismatch.
- Points deviating in tails indicate heavy or light tails.

QQ plot (continued)

To simplify this

- Let $x_1, x_2, \dots, x_n$ represent n observations on a characteristic X_i .
- Let $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$ denote these observations ordered by magnitude.
- The values $x_{(j)}$ are called sample quantiles, where $x_{(1)}$ is the smallest and $x_{(n)}$ is the largest.
- When all $x_{(j)}$ are distinct, exactly j observations are less than or equal to $x_{(j)}$.
- The proportion of observations less than or equal to $x_{(j)}$ is approximately $(j - \frac{1}{2}) / n$.
- For a standard normal distribution, quantiles $q(j)$ satisfy

$$P[Z \leq q(j)] = \int_{-\infty}^{q(j)} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = p(j) = \frac{j - \frac{1}{2}}{n}.$$

- Here $p(j)$ is the probability of obtaining a value less than or equal to $q(j)$ from a standard normal draw.
- The idea of the QQ diagram is to look at pairs $(q(j), x_{(j)})$ with the same cumulative probability $(j - \frac{1}{2}) / n$.
- If the data follow a normal population, the points $(q(j), x_{(j)})$ lie approximately on a straight line.
- This occurs because $\sigma q(j) + \mu$ is nearly the expected sample quantile.

Example

A sample of $n = 10$ observations is given. The ordered observations $x_{(j)}$, the probability levels $(j - \frac{1}{2})/n$, and the corresponding standard normal quantiles $q(j)$ are shown below.

Ordered observations $x_{(j)}$	$(j - \frac{1}{2})/n$	Standard normal quantiles $q(j)$
-1.00	0.05	-1.645
-0.10	0.15	-1.036
0.16	0.25	-0.674
0.41	0.35	-0.385
0.62	0.45	-0.125
0.85	0.55	0.125
1.26	0.65	0.385
1.54	0.75	0.674
1.71	0.85	1.036
2.30	0.95	1.645

Example (continued)

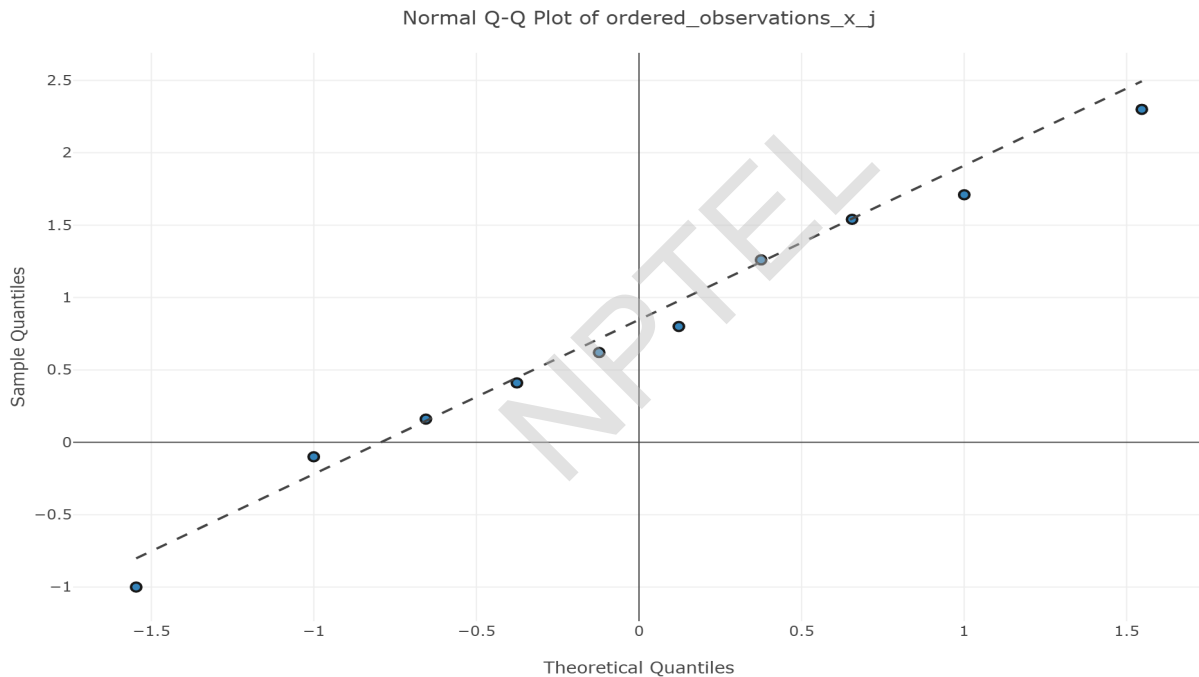
For example,

$$P[Z \leq 0.385] = \int_{-\infty}^{0.385} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 0.65.$$

This matches the probability level $(j - \frac{1}{2})/n$.

The Q-Q plot constructed from these values plots $x_{(j)}$ against $q(j)$. The points lie close to a straight line, indicating that the sample is consistent with a normal population, even with $n = 10$.

Plot



Maximum Likelihood Estimate for Multivariate Normal Distribution

Let us assume that the $p \times 1$ vectors $X_1, X_2, \dots, X_n$ represent a random sample from a multivariate normal population with mean vector μ and covariance matrix Σ . Since $X_1, X_2, \dots, X_n$ are mutually independent and each has distribution $\mathcal{N}_p(\mu, \Sigma)$, the joint density function of all the observations is the product of the marginal normal densities:

$$\begin{aligned}\{\text{Joint density of } X_1, X_2, \dots, X_n\} &= \prod_{j=1}^n \left[\frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} e^{-(x_j - \mu)^\top \Sigma^{-1} (x_j - \mu)/2} \right] \\ &= \frac{1}{(2\pi)^{np/2} |\Sigma|^{n/2}} e^{-\frac{1}{2} \sum_{j=1}^n (x_j - \mu)^\top \Sigma^{-1} (x_j - \mu)}\end{aligned}$$

When the numerical values of the observations become available, they may be substituted for the x_j in the above equation. The resulting expression, now considered as a function of μ and Σ for the fixed set of observations $x_1, x_2, \dots, x_n$, is called the likelihood.

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

Now the exponent in the joint density can be simplified.

$$(x_j - \mu)^\top \Sigma^{-1} (x_j - \mu) = \text{tr}[(x_j - \mu)^\top \Sigma^{-1} (x_j - \mu)] = \text{tr}[\Sigma^{-1} (x_j - \mu)(x_j - \mu)^\top]$$

Next,

$$\sum_{j=1}^n (x_j - \mu)^\top \Sigma^{-1} (x_j - \mu) = \sum_{j=1}^n \text{tr}[(x_j - \mu)^\top \Sigma^{-1} (x_j - \mu)] = \sum_{j=1}^n \text{tr}[\Sigma^{-1} (x_j - \mu)(x_j - \mu)^\top] = \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \mu)(x_j - \mu)^\top \right) \right]$$

Since the trace of a sum of matrices is equal to the sum of the traces of the matrices.

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

We can add and subtract $\bar{x} = (1/n) \sum_{j=1}^n x_j$ in each term $(x_j - \mu)$ in $\sum_{j=1}^n (x_j - \mu)(x_j - \mu)^\top$ to give

$$\sum_{j=1}^n (x_j - \bar{x} + \bar{x} - \mu)(x_j - \bar{x} + \bar{x} - \mu)^\top = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + \sum_{j=1}^n (\bar{x} - \mu)(\bar{x} - \mu)^\top = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + n(\bar{x} - \mu)(\bar{x} - \mu)^\top$$

because the cross-product terms, $[\sum_{j=1}^n (x_j - \bar{x})(\bar{x} - \mu)^\top$ and $\sum_{j=1}^n (\bar{x} - \mu)(x_j - \bar{x})^\top$, are both matrices of zeros.

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

Consequently, using Equations above, we can write the joint density of a random sample from a multivariate normal population as

$$\begin{aligned} & \{ \text{Joint density of } X_1, X_2, \dots, X_n \} \\ &= (2\pi)^{-np/2} |\Sigma|^{-n/2} \\ & \times \exp \left\{ -\text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + n(\bar{x} - \mu)(\bar{x} - \mu)^\top \right) \right] / 2 \right\} \end{aligned}$$

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

Substituting the observed values $x_1, x_2, \dots, x_n$ into the joint density yields the likelihood function. We shall denote this function by $L(\mu, \Sigma)$, to stress the fact that it is a function of the (unknown) population parameters μ and Σ . Thus, when the vectors x_j contain the specific numbers actually observed, we have

$$L(\mu, \Sigma) = \frac{1}{(2\pi)^{np/2} |\Sigma|^{n/2}} \exp \left\{ -\frac{1}{2} \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + n(\bar{x} - \mu)(\bar{x} - \mu)^\top \right) \right] \right\}$$

It will be convenient to use the exponent in the likelihood function in different ways. In particular, we shall make use of the identity

Maximum Likelihood Estimate for Multivariate Normal Distribution (continued)

$$\begin{aligned} & \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top + n(\bar{x} - \mu)(\bar{x} - \mu)^\top \right) \right] \\ &= \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top \right) \right] + n \text{tr} [\Sigma^{-1}(\bar{x} - \mu)(\bar{x} - \mu)^\top] \\ &= \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top \right) \right] + n(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu) \end{aligned}$$

Theorem

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal population with mean μ and covariance Σ . Then

$$\hat{\mu} = \bar{X} \quad \text{and} \quad \hat{\Sigma} = \frac{1}{n} \sum_{j=1}^n (X_j - \bar{X})(X_j - \bar{X})^\top = \frac{(n-1)}{n} S$$

are the maximum likelihood estimators of μ and Σ , respectively. Their observed values, $\bar{x}$ and $(1/n) \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top$, are called the maximum likelihood estimates of μ and Σ .

Theorem (Proof)

Proof. The exponent in the likelihood function, apart from the multiplicative factor $-\frac{1}{2}$, is

$$\text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top \right) \right] + n(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu)$$

Σ^{-1} is positive definite, so the distance $(\bar{x} - \mu)^\top \Sigma^{-1}(\bar{x} - \mu) > 0$ unless $\mu = \bar{x}$. Thus, the likelihood is maximized with respect to μ at $\hat{\mu} = \bar{x}$. It remains to maximize

$$L(\mu, \Sigma) = \frac{1}{(2\pi)^{np/2} |\Sigma|^{n/2}} \exp \left\{ -\frac{1}{2} \text{tr} \left[\Sigma^{-1} \left(\sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top \right) \right] \right\}$$

over Σ .

Theorem (Proof)

By the following result:

Given a $p \times p$ symmetric positive definite matrix B and a scalar $b > 0$,

$$\frac{1}{|\Sigma|^b} e^{-\text{tr}(\Sigma^{-1}B)/2} \leq \frac{1}{B^b} (2b)^{pb} e^{-bp}$$

for all positive definite $\Sigma_{p \times p}$, with equality holding only for $\Sigma = (1/2b)B$.

with $b = n/2$ and

$$B = \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top,$$

the maximum occurs at

$$\hat{\Sigma} = \frac{1}{n} \sum_{j=1}^n (x_j - \bar{x})(x_j - \bar{x})^\top.$$

The maximum likelihood estimators are random quantities. They are obtained by replacing the observations $x_1, x_2, \dots, x_n$ in the expressions for $\hat{\mu}$ and $\hat{\Sigma}$ with the corresponding random vectors $X_1, X_2, \dots, X_n$.

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Thank you

